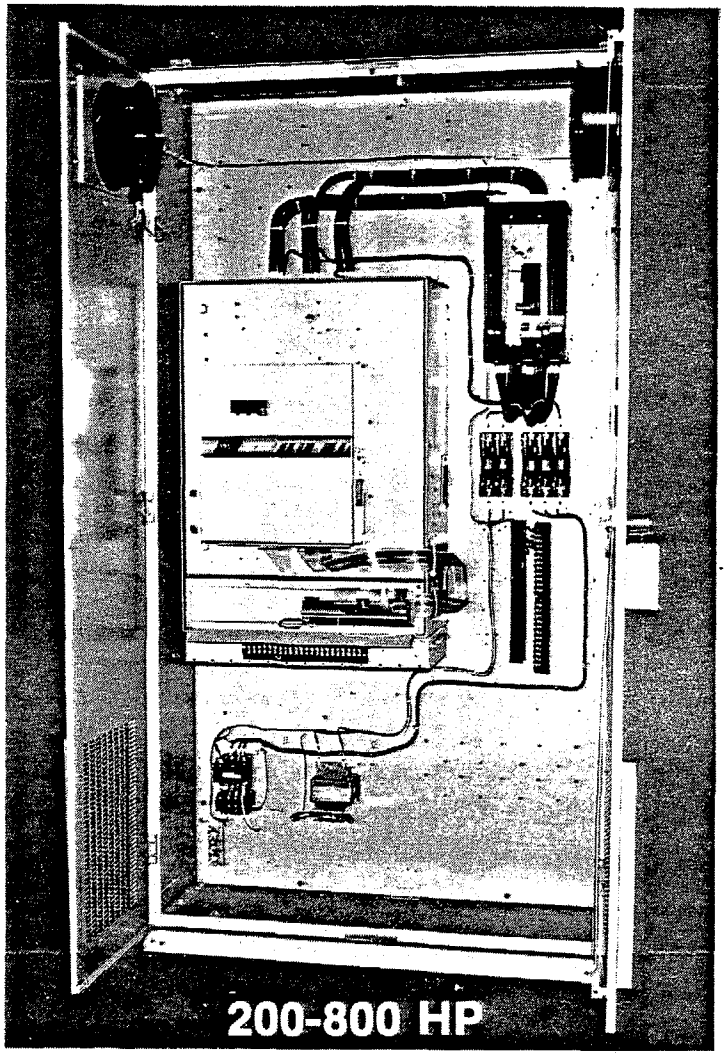
GEK-85771

GE Drives

INSTRUCTIONS



DC-285™ ADJUSTABLE SPEED DRIVES

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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INTRODUCTION

This Instruction Book contains helpful suggestions for placing the DC-285 drive equipment in service. It contains general information about drive operation and maintenance.

The operator and person responsible for maintaining this equipment should at all times have access to a copy of this instruction book and should consult this instruction book prior to operation or maintenance of the equipment.

RECEIVING HANDLING AND STORAGE

RECEIVING

The equipment should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from GENERAL ELECTRIC COMPANY, DRIVE PRODUCTS OPERATION, ERIE, PA. When seeking assistance, please use drive seriql number to identify the equipment.
Telephone (814) 455-3219.

HANDLING

Wall mounted power units can be transported by lift trucks with the forks completely under the base, care being taken that the unit does not tip.

STORAGE

If the equipment is not to be installed immediately, it should be stored in a clean, dry location with ambient temperatures between —20° C (—4° F) and +55° C (131° F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage environment exceeds a 15° C (27° F) drop in temperature at 50% humidity over a 4-hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat.) Higher humidities with smaller temperature changes will also cause condensation.

Condensation occurs when air containing some moisture is cooled below its dew point. The dew point represents saturation of the air, and is the temperature at which the moisture starts to condense into water. It is not a fixed temperature but rather is related to the initial temperature of the air and its relative humidity at that temperature. The amount of moisture that can be held in the air is related to the air temperature. The following example illustrates some of these relationships.

In industrial drives, condensation is a possibility in applications where air temperature changes are large and rapid and/or the air is moist. For example, an outdoor crane operating in sunshine on a winter day, which then is shut down and parked in the shade will experience a rapid drop in temperature. This can result in condensation inside the equipment. Adding heat to keep the air temperature above its dew point can prevent condensation.

If storage temperatures below —20° C (—4° F) are likely to be present then auxiliary heat should be added in each enclosure to maintain temperature at or above —20° C. For assistance in heater size selection, contact General Electric Company.

When a drive that has been in operation is shut down for either a short or extended period of time, it is recommended that the environmental conditions be maintained the same as when in operation. Power unit ventilation or heating and air conditioning (if used) should be left on during the downtime to prevent large changes in temperature and possible moisture condensation.

TABLE I
Relationships Between Air Temperature, Relative Humidity and Dew Point

AIR TEMP °F	AIR TEMP °C	RELATIVE HUMIDITY %	WGT. OF MOISTURE IN 1 LB. OF DRY AIR GRAINS	DEW POINT °F	DEW POINT °C
104	40	100	345	104	40
104	40	80	270	97	36
104	40	40	130	75	24
104	40	10	32	37	3
50	10	100	54	50	10
50	10	80	42	43	6
50	10	40	21	25	4

SAFETY FOR PERSONNEL AND EQUIPMENT

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS, CAUTIONS, AND NOTES AS USED IN THIS PUBLICATION

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

WHEN A WARNING LABEL IS PROVIDED IT WILL BE RED IN COLOR WITH BLACK OR WHITE LETTERING.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF, THE EQUIPMENT.

WHEN A CAUTION LABEL IS PROVIDED IT WILL BE AMBER IN COLOR WITH BLACK LETTERING.

NOTE

NOTES MERELY CALL ATTENTION TO INFORMATION THAT IS ESPECIALLY SIGNIFICANT IN UNDERSTANDING AND OPERATING THE EQUIPMENT.

WHEN A NOTE LABEL IS PROVIDED IT WILL BE WHITE IN COLOR WITH BLACK LETTERING.

WARNINGS

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT. FOLLOW THE START-UP PROCEDURES STEP BY STEP.

READ AND HEED ALL **WARNING, CAUTION AND NOTE** LABELS POSTED ON THE EQUIPMENT.

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY.
LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

WARNINGS

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS.

WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT GREAT CAUTION MUST BE USED. THE OSCILLOSCOPE'S CHASSIS SHOULD BE GROUNDED AND A DIFFERENTIAL AMPLIFIER INPUT SHOULD BE USED. CARE SHOULD BE USED IN THE SELECTION OF PROBES AND LEADS AND IN THE ADJUSTMENT OF THE OSCILLOSCOPE SO THAT ACCURATE READINGS MAY BE MADE. SEE OSCILLOSCOPE MANUFACTURERS INSTRUCTION BOOK FOR PROPER OPERATION AND ADJUSTMENTS TO THE OSCILLOSCOPE.

SOME CONTROLLERS ARE FURNISHED WITH PARTIAL ENCLOSURES OPEN AT TOP AND BOTTOM. THESE ARE INTENDED ONLY FOR MOUNTING IN ANOTHER ENCLOSURE OR IN A CONTROL ROOM HAVING ACCESS BY QUALIFIED PERSONNEL ONLY.

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE CONTROLLERS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE FLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE CONTROLLERS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH DC MOTORS SUITABLE FOR USE IN THESE LOCATIONS.

ALL MOTOR BASES AND EQUIPMENT ENCLOSURE HOUSINGS SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEM.

CAUTIONS

DO NOT REMOVE CONNECTIONS OR PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

DO NOT REMOVE INPUT POWER FROM THE DRIVE UNTIL IT HAS FULLY EXECUTED A STOP SEQUENCE AS THIS CAN DAMAGE THE DRIVE SYSTEM.

INSTALLATION WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND BE CONSISTENT WITH ALL LOCAL CODES. SECONDARIES OF 115 VOLT CONTROL TRANSFORMERS TYPICALLY HAVE ONE SIDE FUSED AND THE OTHER GROUNDED OR AVAILABLE FOR GROUNDING BY THE USER.

THE ELEMENTARY DIAGRAMS SUPPLIED WITH THE DC 285 DRIVE CONTROLLER IDENTIFY SOME SIGNAL CONNECTIONS AS "LOW LEVEL ANALOG AND/OR DIGITAL SIGNALS" AND INDICATE SPECIAL WIRING PRACTICES. THESE SIGNALS SHOULD BE ISOLATED FROM ALL OTHER WIRING (PARTICULARLY ANY POWER WIRES) AND RUN IN A SEPARATE CONDUIT OR WIRE RUN TO PREVENT POSSIBLE MISOPERATION OF THE CONTROLLER

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HIPOT WITHOUT CONSULTING THE DRIVE PRODUCTS OPERATION. GENERAL ELECTRIC COMPANY.

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN THOSE SHOWN ON THE ELEMENTARY DIAGRAM. CONNECTION OF DEVICES SUCH AS VOLTMETERS ON THE TACHOMETER MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING THE DRIVE. PRODUCTS OPERATION, GENERAL ELECTRIC COMPANY. DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

NOTES

IT IS RECOMMENDED THAT THE DRIVE SYSTEM COMMON (⏚) CIRCUIT BE GROUNDED (⏚) AT ONLY ONE POINT. THIS MEANS THAT IF THE DRIVE REFERENCE IS SUPPLIED BY A NUMERICAL REFERENCE IS SUPPLIED BY A NUMERICAL CONTROL OR PROCESS INSTRUMENT WITH GROUNDED COMMON, THE DRIVE COMMON SHOULD NOT BE GROUNDED. IF A SEPARATE POWER TRANSFORMER IS USED AND IF THE SECONDARY OF THE TRANSFORMER MUST BE GROUNDED, IT IS RECOMMENDED THAT HIGH RESISTANCE GROUNDING BE USED.

INSTALLATION

LOCATION

DC-SCR drive power units are suitable for most factory areas where other industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90 percent. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dirt, dust or foreign matter
2. Vibration or shock
3. Moisture or vapors
4. Temperature excursions
5. Caustic fumes
6. Power line fluctuations
7. Electromagnetic interference or "**NOISE**" introduced by —

    a. Radio frequency signals, typically portable transmitters used in the vicinity of the equipment or its wiring.

    b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay/contactor/brake coils as a part of or in the vicinity of drive control circuits.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways, for signal, control and power wiring levels defined as follows:

   a) Signal — Speed and/or position signals power supplies and logic signals (see system elementaries and notes to identify wire levels).

   b) Control — AC or DC control circuits

   c) Power — Armature and field leads Brakes, 115 VAC circuits

2) Signal wiring and power wiring may cross at right angles with a minimum one inch separation. Avoid parallel runs between signal and power wiring harnesses. If signal wire harnesses must be run in parallel to control or power wire harnesses, a minimum of a four-inch separation must be maintained.

GROUNDING

1) Drive common (PCOMX) should be grounded at **only one** point. If the reference is supplied by numerical control or process instrument with a grounded common the drive common should not be grounded separately.

2) If an isolation transformer is used and must be grounded, it is recommended that a high resistance ground be used.

3) Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

COMMONS

Connection of reference, meter or any other external connection of commons can only be made at 3TB—PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

1) Incoming AC line connections.

2) Power connections to motor armature and motor field.

3) All terminal board (TB) connections including: 3TB, 4TB and 6TB.

MOUNTING

Totally enclosed controllers should be positioned to permit head radiation from all surfaces.

A wall mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the door may be fully opened for easy access.

Wall mounted controllers may be mounted on any firm, reasonable flat, vertical surface.

CONNECTIONS

All internal electrical connections between components in the basic DC-SCR drive power units are made at General Electric Company. Various modification options may be purchased, mounted and wired by the user or authorized GE distributor.

Be sure to protect the interior panel mounted components and sub-assemblies from metal particles when cutting or drilling entrances for interconnecting wiring and cables.

NOTE

OCCASIONALLY DRIVE ISOLATION TRANSFORMERS AND/OR OTHER ELECTRICAL COMPONENTS ARE SHIPPED SEPARATELY AND MUST BE MOUNTED AND CONNECTED TO THE SYSTEM.

NOTE

CONNECTION OF EXTERNAL CIRCUITS OTHER THAN SHOWN ON THE ELEMENTARY DIAGRAM, SUCH AS VOLTMETERS ON THE TACHOMETER, MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

Before power is applied to the drive system, checks should be made to see that all internal connections are tight, and that all open relays and contactors operate freely by hand. Check that the equipment is clean and that no metal chips are present.

MAINTENANCE

Periodically inspect and maintain the equipment protective devices (particularly air filters when supplied) per instructions in this section. Check all electrical connections for tightness; look for signs of poor connections and overheating (arcing or discoloration).

DC MOTORS

Maintenance instructions covering brushes, commutator and lubrication are in GEH-2304 or GEH-3967 which are provided separately.

CAUTION

IT SHOULD BE NOTED THAT IF THE AC INPUT POWER IS NOT REMOVED FROM THE POWER UNIT THE MOTOR FIELD WILL CONTINUE TO BE EXCITED EVEN AT STAND STILL WITH THE MD CONTACTOR DROPPED OUT.

ALTHOUGH DC MOTORS ARE DESIGNED FOR FULL FIELD EXCITATION AT STAND STILL FOR PERIODS OF TIME (SEVERAL HOURS), UNDER THIS CONDITION THE MD CONTACTOR SHOULD BE DROPPED OUT, OTHERWISE THE FIELD COILS WILL BE SUBJECTED TO EXCESSIVE TEMPERATURE AND SIGNIFICANTLY REDUCED INSULATION LIFE.

WHENEVER THE MD CONTACTOR IS DROPPED OUT, A FIELD ECONOMY CIRCUIT WHICH AUTOMATICALLY REDUCES THE LEVEL OF EXCITATION IS EMPLOYED.

PRINTED CIRCUIT CARDS

Printed circuit cards normally do not require maintenance except to keep them clean and tightly secured. Clean as follows:

1. **Dry Dust** — Vacuum clean, then blow with dry filtered compressed air (low pressure supply).

2. **Oily Dirt** — Certain components (electrolytic capacitors, switches, meters, potentiometers and transformers) can be damaged by solvent, so use is not recommended. If absolutely necessary, clean using solvent sparingly and avoid above components. Clean contact terminals with dry non-linting cloth after solvent has been used. Recommended solvents: Freon TE or TF.

3. If the card is badly contaminated or corroded, replace.

SILICON CONTROLLED RECTIFIERS

Keep SCRs and heatsink free from dirt, oil or grease, since any accumulation of dirt may cause overheating. Clean as follows:

1. **Dry Dust** — Vacuum clean, then blow with dry, filtered compressed air (low pressure).

CAUTION

SOLVENT CAN HARM NON-METAL COMPONENTS.

2. **Oily Dirt** — Use dry or barely moist (with solvent) non-linting cloth. Repeat until cloth remains clean. All SCRs must be cleaned with dry non-linting cloth after solvent has been used. Recommended solvents: Freon TE or TF.

CONTROL DEVICES

Inspect all relays and contactors at regular intervals and keep them free from dirt, oil or grease. Check for freedom of moving parts, corrosion, loose connections, worn or broken parts, charred insulation or odor, proper contact pressure and remaining wear allowance on contacts. Do not lubricate the contacts as lubrication shortens their life.

Both copper and silver contacts will become darkened and somewhat roughened in normal operation. This does not interfere with their performance, and does not indicate that the contacts should be filed. In general, contacts will not need attention during their normal life, but if prominent beads form on the surfaces due to severe arcing, the contact faces may be dressed with a fine file. **Do not** use sandpaper or emery cloth.

Any contact that is worn to the point where contact wipe or pressure is lost should be replaced. Cleaning procedure is the same as previously given for SCR heatsink.

GENERAL DESCRIPTION (Figure 2)

Power is fed through the AC fuses through current transformers and enters the power conversion SCR modules where it is converted to DC adjustable voltage. DC current is fed through a DC fuse (four quadrant controllers only) and MD contactor to the DC motor armature.

The basic elements of the DC-285 drive controller are shown in the simplified block diagram, Figure 4.

The speed of the motor is proportional to the DC voltage applied to its armature. Speed is measured and controlled by motor CEMF feedback (armature voltage feedback with **IR** compensation). As an optional feature, speed can be measured and regulated by a DC or AC tachometer feedback.

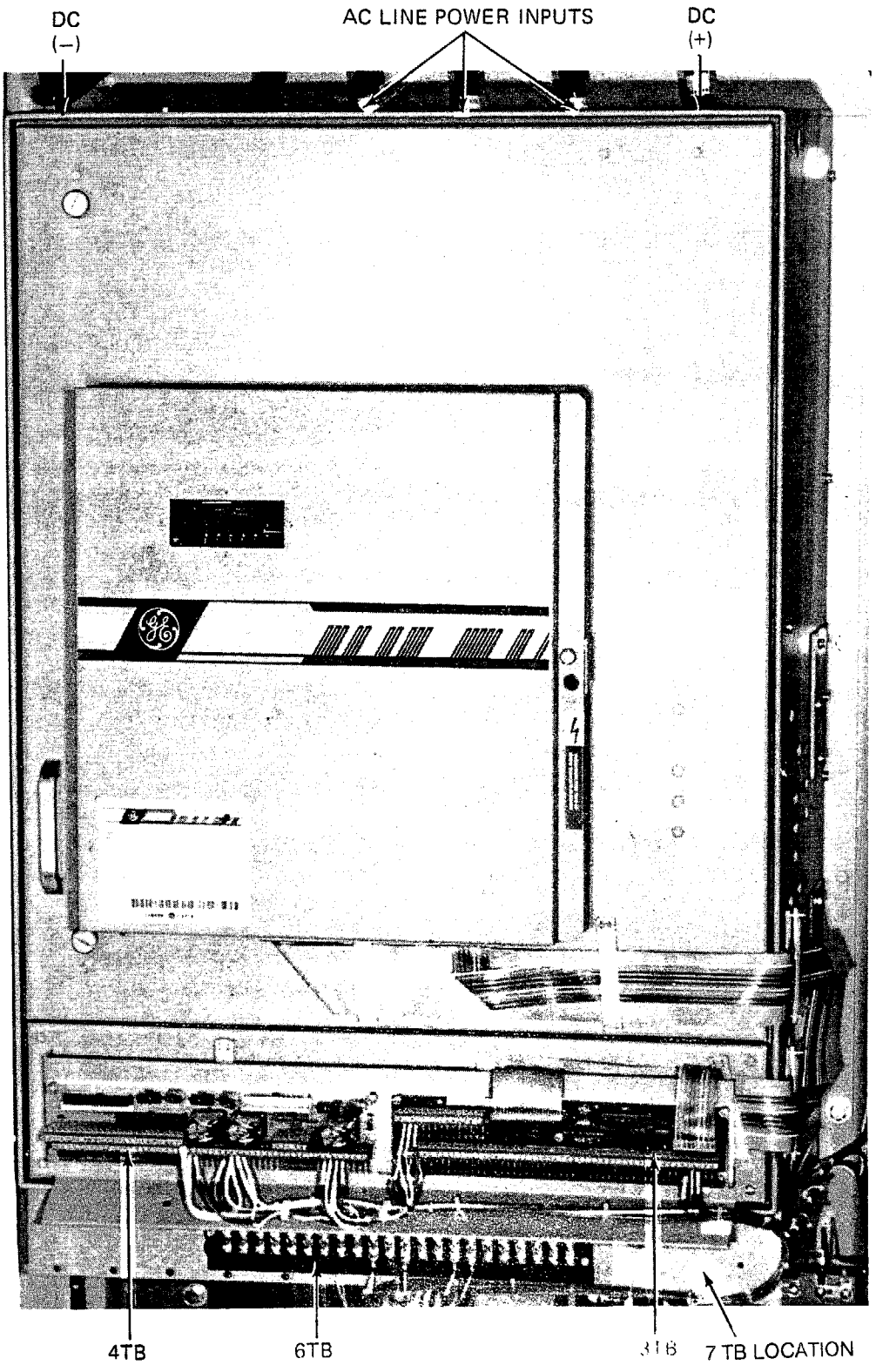


FIGURE 1
DC-285 CONTROLLER SHOWING
CONTROL AND POWER CONNECTIONS

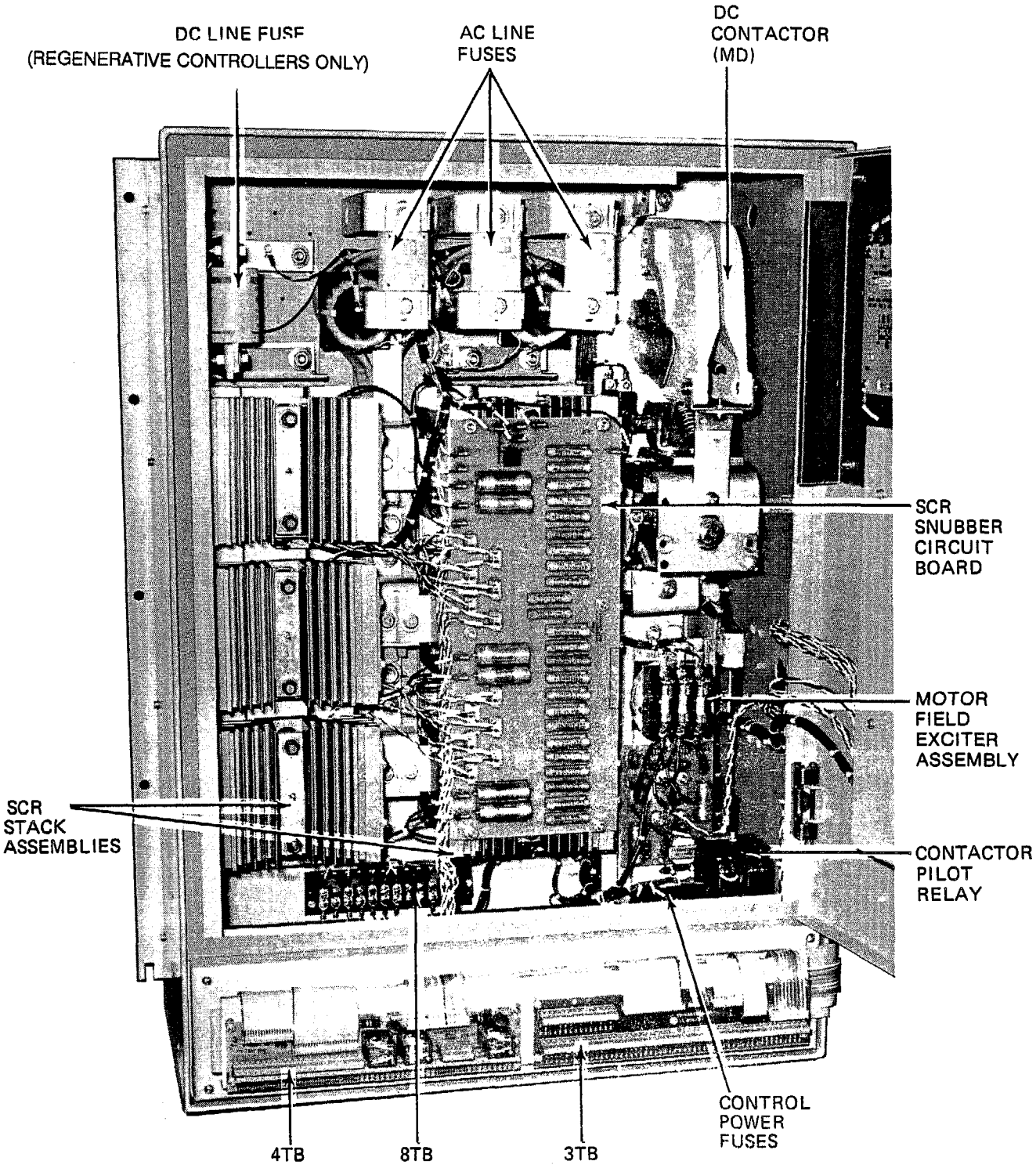


FIGURE 2
DC-285 POWER CONTROLLER ASSEMBLY

TABLE II
POWER RATINGS — 3 PHASE 50/60HZ

	AMPS - - INPUT/OUTPUT											
	230 VAC			460 VAC			575 VAC					
		OUTPUT			OUTPUT			OUTPUT			OUTPUT	
HP	INPUT LINE	240VDC ARM	150VDC FIELD	INPUT LINE	500VDC ARM	300VDC FIELD	INPUT LINE	630VDC ARM	300VDC FIELD	INPUT LINE	700VDC ARM	300VDC FIELD
125	359	440	24	—	—	—	—	—	—	—	—	—
150	428	525	24	—	—	—	—	—	—	—	—	—
200	543	666	24	—	—	—	—	—	—	—	—	—
250	698	855	24	—	—	—	—	—	—	—	—	—
300	837	1025	24	412	505	24	—	—	—	—	—	—
400	—	—	—	547	670	24	430	530	24	390	480	24
500	—	—	—	677	830	24	540	660	24	490	600	24
600	—	—	—	804	985	24	645	790	24	580	710	24
700	—	—	—	934	1145	24	735	900	24	661	810	24
800	—	—	—	1057	1285	24	850	1040	24	763	935	24
900	—	—	—	—	—	—	945	1160	24	847	1040	24
1000	—	—	—	—	—	—	1040	1280	24	938	1150	24

SERVICE CONDITIONS

- Rated Voltage — 230 VAC (-5, + 10%)
460 VAC (-5, + 10%)
575 VAC (-5, + 10%)
- Frequency — 60 Hertz (± 2 Hertz)
50 Hertz (± 2 Hertz)
- Ambient Temperature — 0° to 40°C (32° to 104°F)
- Altitude — Sea Level to 3300 feet
- Load Inertia — Not more than motor inertia when referred to motor shaft. For motor WK^2 refer to motor sheet.

CONTROLLED SPEED RANGE: 10:1

TABLE III SPEED VS TORQUE:
Typical data for speed versus torque:

	Percent Rated Speed			
	100-60	50	40	5
MOTOR ENCLOSURE	Percent Continuous Torque (Or Rated Amperes)			
DPFG (SELF VENT)	100	94	87	45
TENV or TEFC (SELF VENT)	100	95	92	77

MOTOR CONTROLLER ELEMENTS AND THEIR FUNCTION

[See Connection Diagrams (Figures 9, 10)
As well as Information Labels (Figures 11, 12)

AC LINE FUSES

AC line fuses (with or without an optional disconnect switch) are provided in the three incoming power lines to the controller and protect the SCRs, the wiring inside the controller and the wiring of the DC output.

FERRITE CORE ASSEMBLY (REGENERATIVE CONTROLLERS ONLY)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to excessive rate of transient voltage change.

MAIN CONTACTOR (MD)

This contactor picks up when running mode is selected and no fault exists in the drive. This contactor drops out under the following conditions:

1. When a "Fault" condition occurs in the drive.
2. When the "Stop" command is selected.
3. When the drive regenerates to near zero speed after the "Stop" command (regenerative controllers only).

DC LINE FUSE* (REGENERATIVE CONTROLLERS ONLY)

A DC line fuse is provided in the DC output line to the motor armature and protects the SCRs and motor from overcurrent sourced from the motor.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuse located on the Control card. The secondary winding provides three isolated voltages:

1. 115 VAC to operate the coil of the main (MD) contactor and controller cooling fans.
   Do not use for external 115V excitation.
2. 38 VAC center-tapped for the Control card to provide DC control voltages necessary for proper controller operation.
   Do not use for external 38V excitation.
3. 16 VAC center-tapped for Control card to provide DC control voltages necessary for proper controller operation.
   Do not use for external 16V excitation.

CURRENT TRANSFORMER ASM (CT)

The current transformer assembly is used to provide armature current feedback signals to the control circuit.

MOTOR FIELD EXCITER ASSEMBLY

The motor field exciter assembly is used to provide excitation to the motor shunt field. The assembly contains the field power module (mounted on a heatsink), fuses and current transformer for the field circuit (see figures 6, 7 and 8). A field current suppression printed circuit board is also part of the assembly. Current feedback is connected to the control card by harness FCPL.

CONTROL CARD (See Figure 13)

This card provides the system intelligence for the drive controller. It contains one microprocessor and performs regulating, sequencing, SCR "burst" firing, diagnostic and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented either in hardware and/or software. An additional function of the card is to display the fault status of the drive by means of 6 diagnostic LEDs mounted on the card. The LEDs which are illuminated are interpreted as a binary number indicative of the fault. See the troubleshooting section for a listing of the Fault codes.

This card provides the necessary ±15V, +5V and ±24V DC power used throughout the controller. This card also contains fuses for the power supply circuitry.

Three relays, "MAX" (K1), SLD1 (K3) and SLDM (K2) are on this card.

POWER CONNECTION CARD (See Figures 16 & 17)

This card contains SCR and DC output resistor-capacitor snubbers and the pulse transformers for the SCRs.

The DC voltage feedback isolation resistors are also mounted on this card.

SNUBBER CARD (See Figure 5)

The snubber card contains SCR resistor-capacitor (RC) snubber networks and the rectifier circuit for supplying DC coil power to the MD contactor.

***NOTE:** On Regenerative Controllers with paralleled DC fuses, replace both fuses, even if only one is blown.

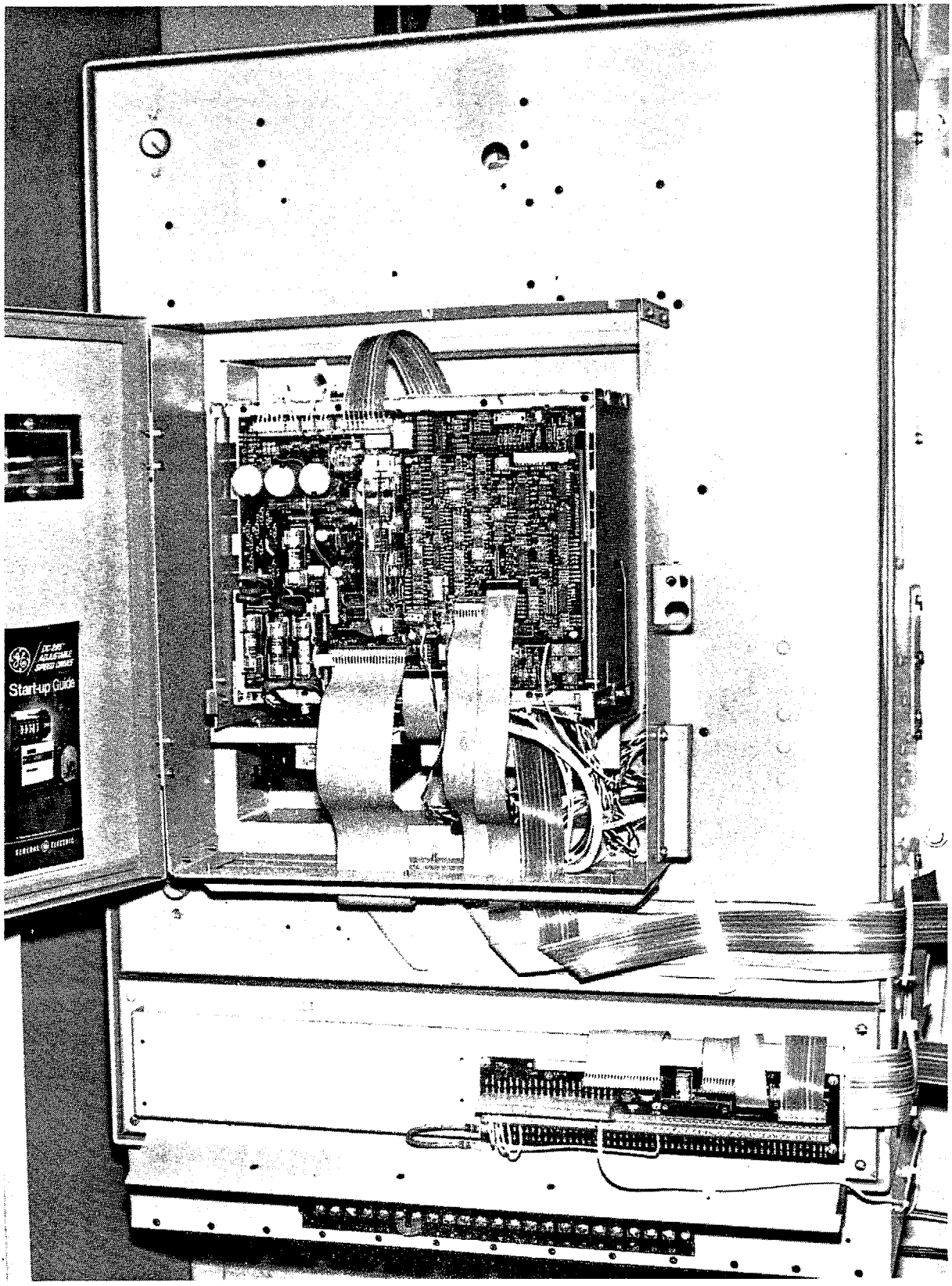


FIGURE 3
DC-285 DRIVE CONTROLLER
WITH DRIVER/REGULATOR

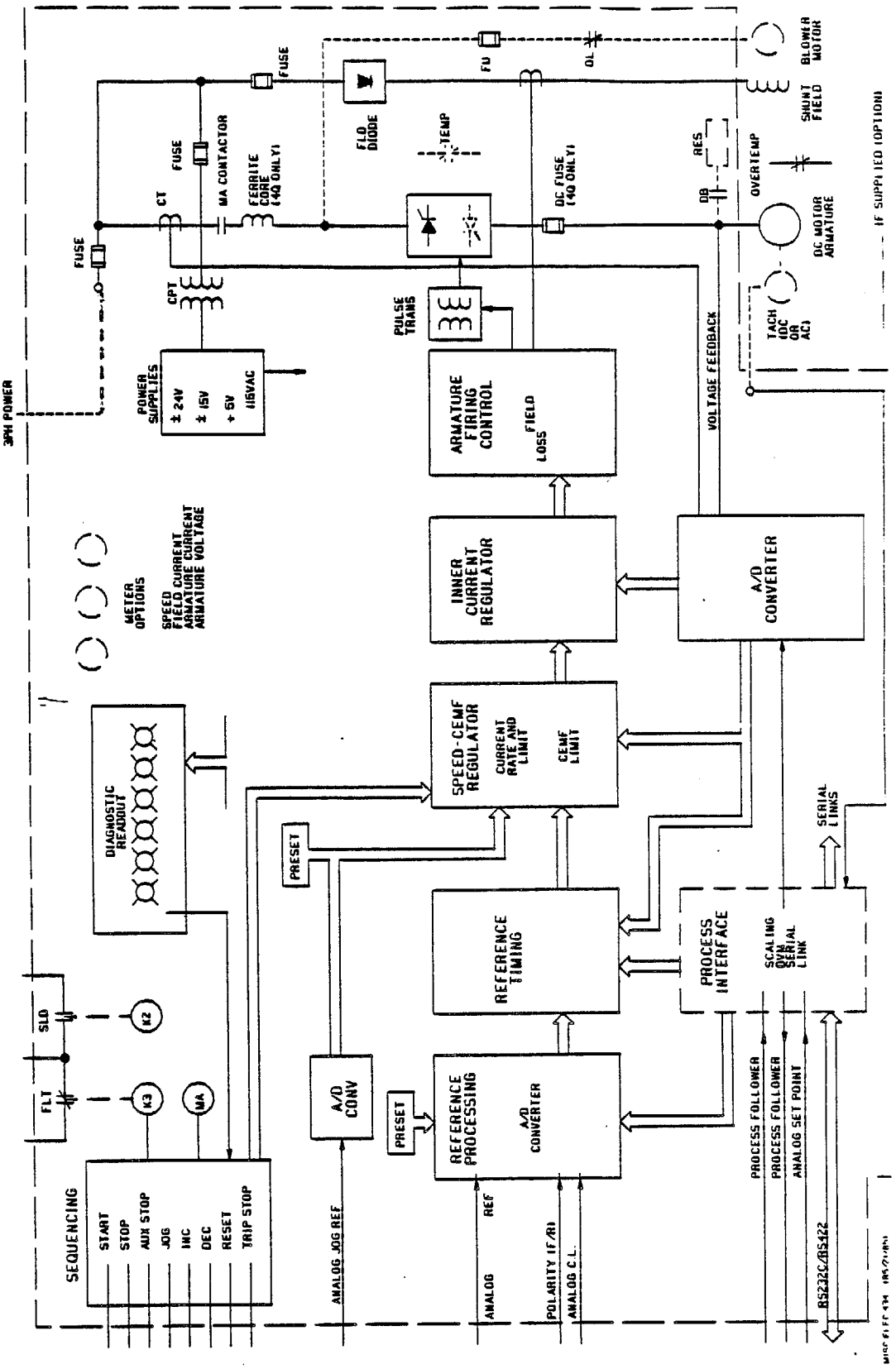


FIGURE 4
BLOCK DIAGRAM -- DC-285 DRIVE CONTROLLER

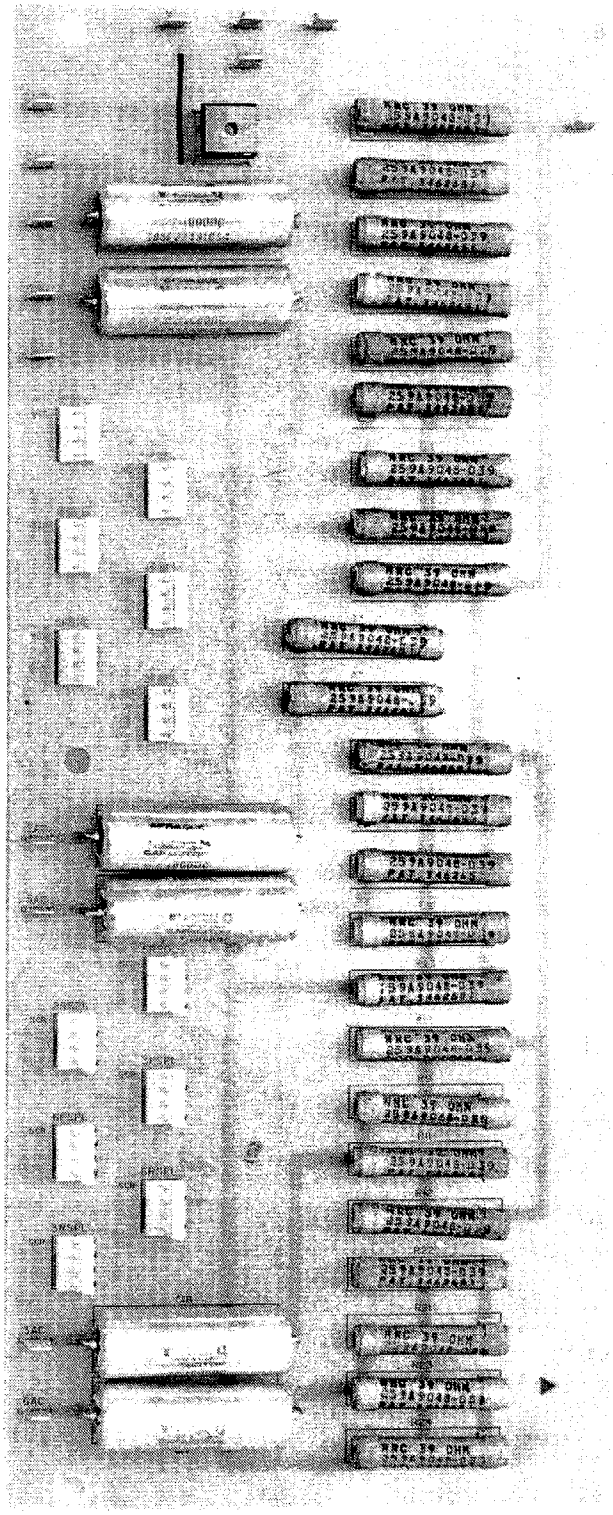


FIGURE 5
DC-285 SNUBBER CARD

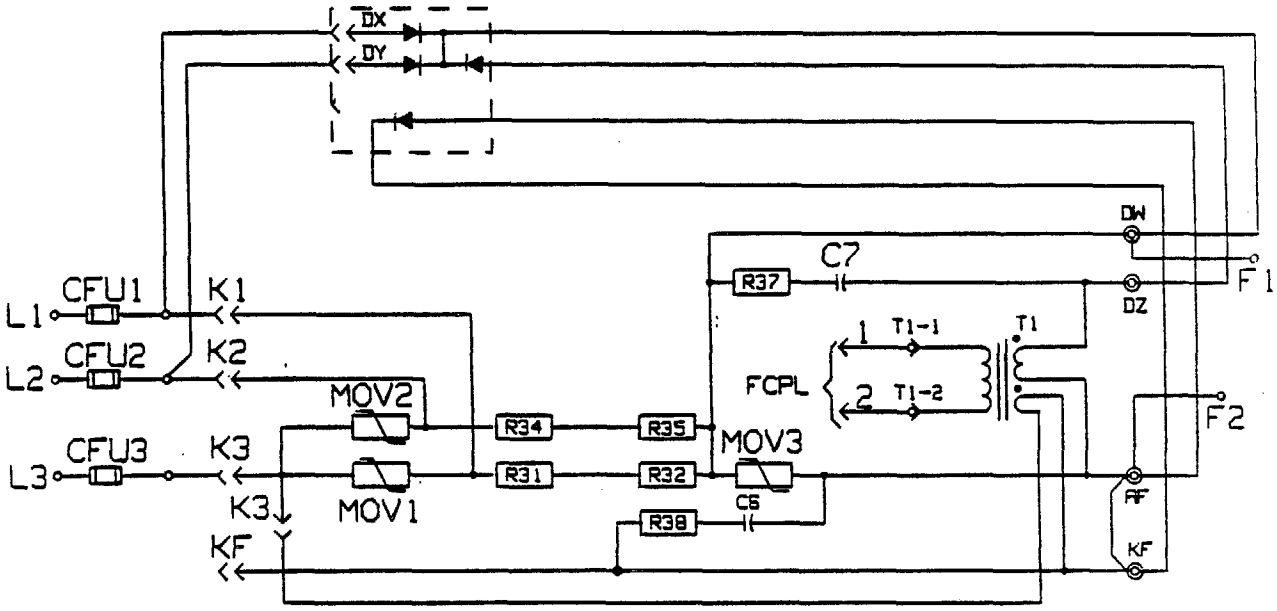


FIGURE 6
MFE POWER CIRCUIT

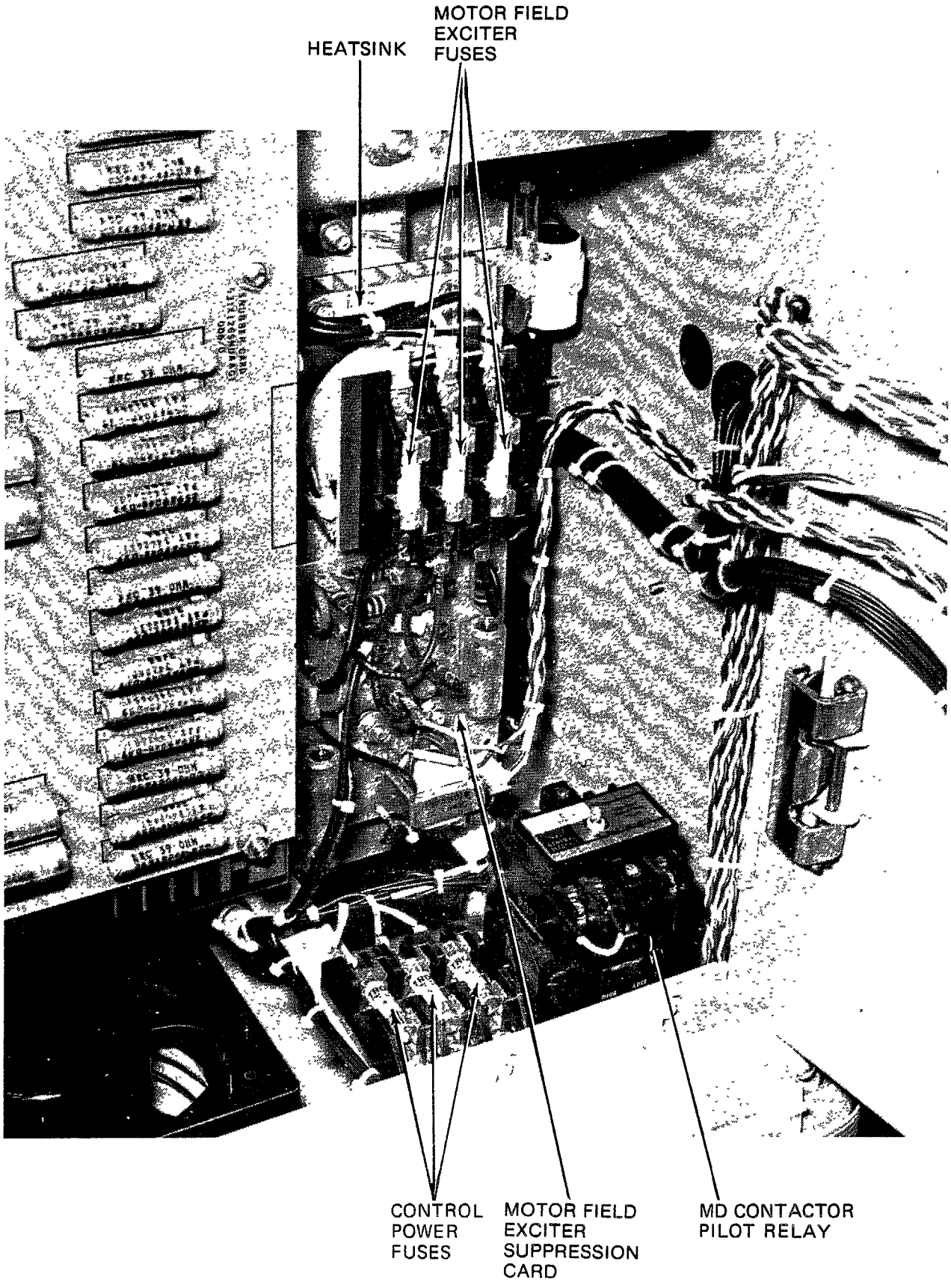


FIGURE 7
MOTOR FIELD EXCITER ASSEMBLY

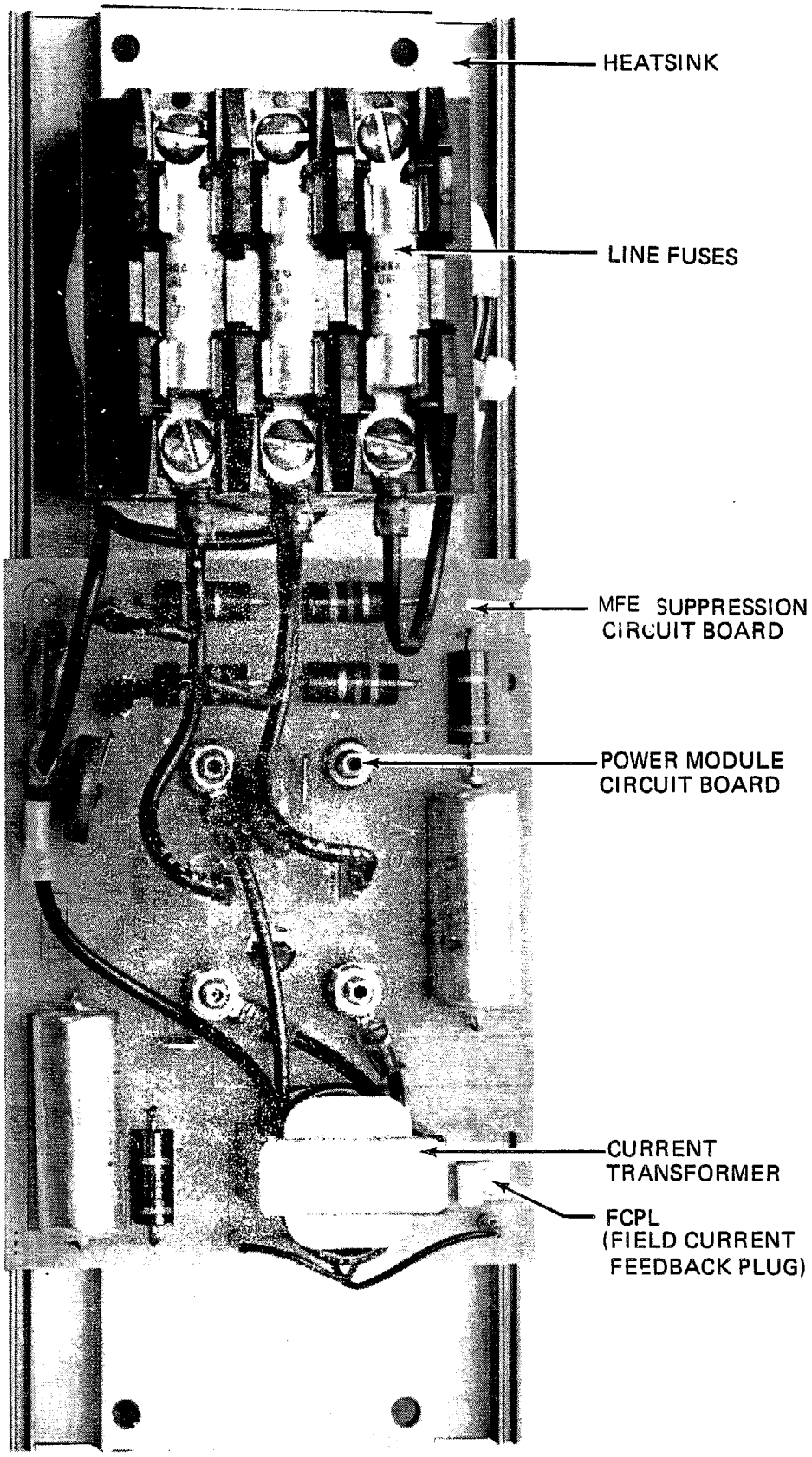


FIGURE 8
MOTOR FIELD EXCITER

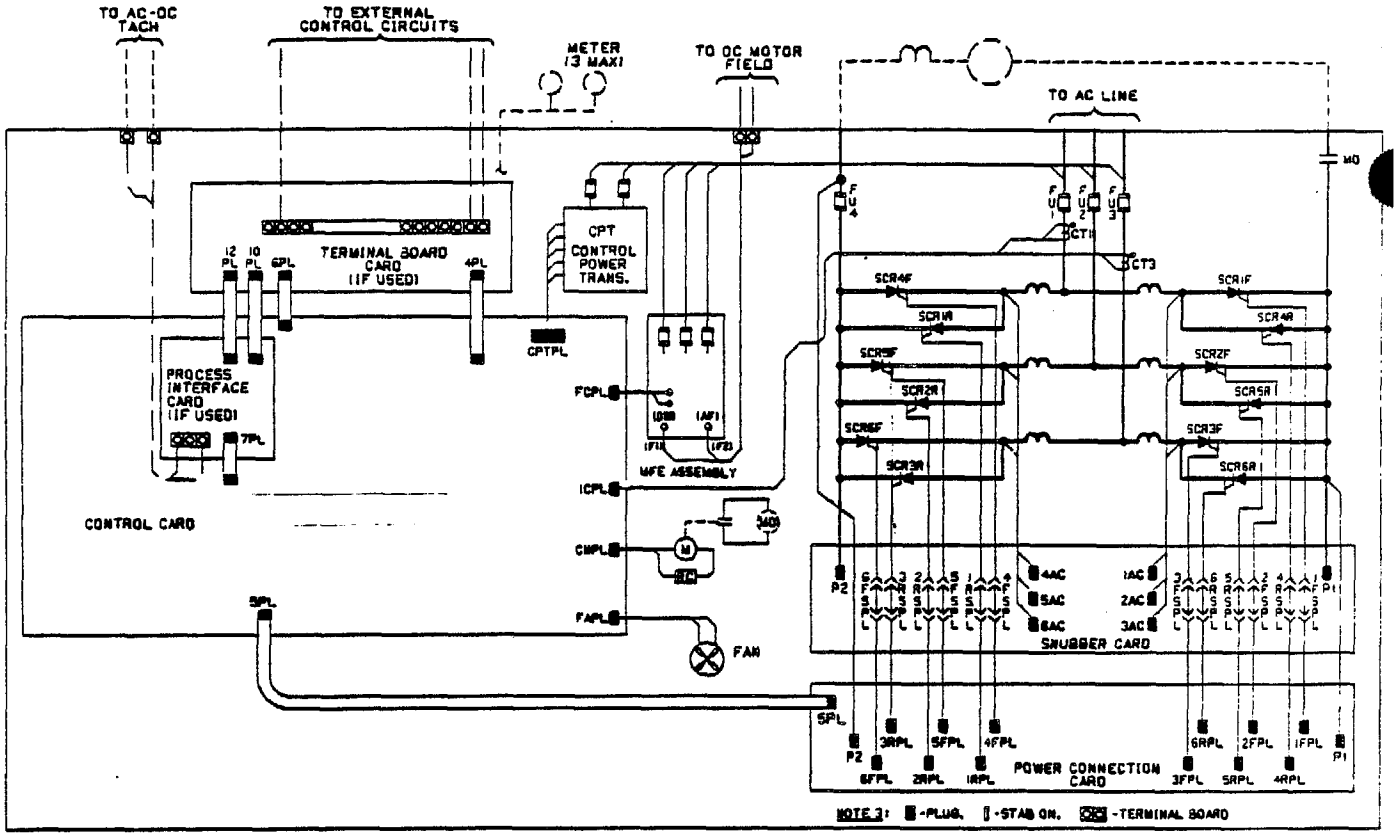


FIGURE 9 — CONNECTION DIAGRAM — REGENERATIVE CONTROLLER

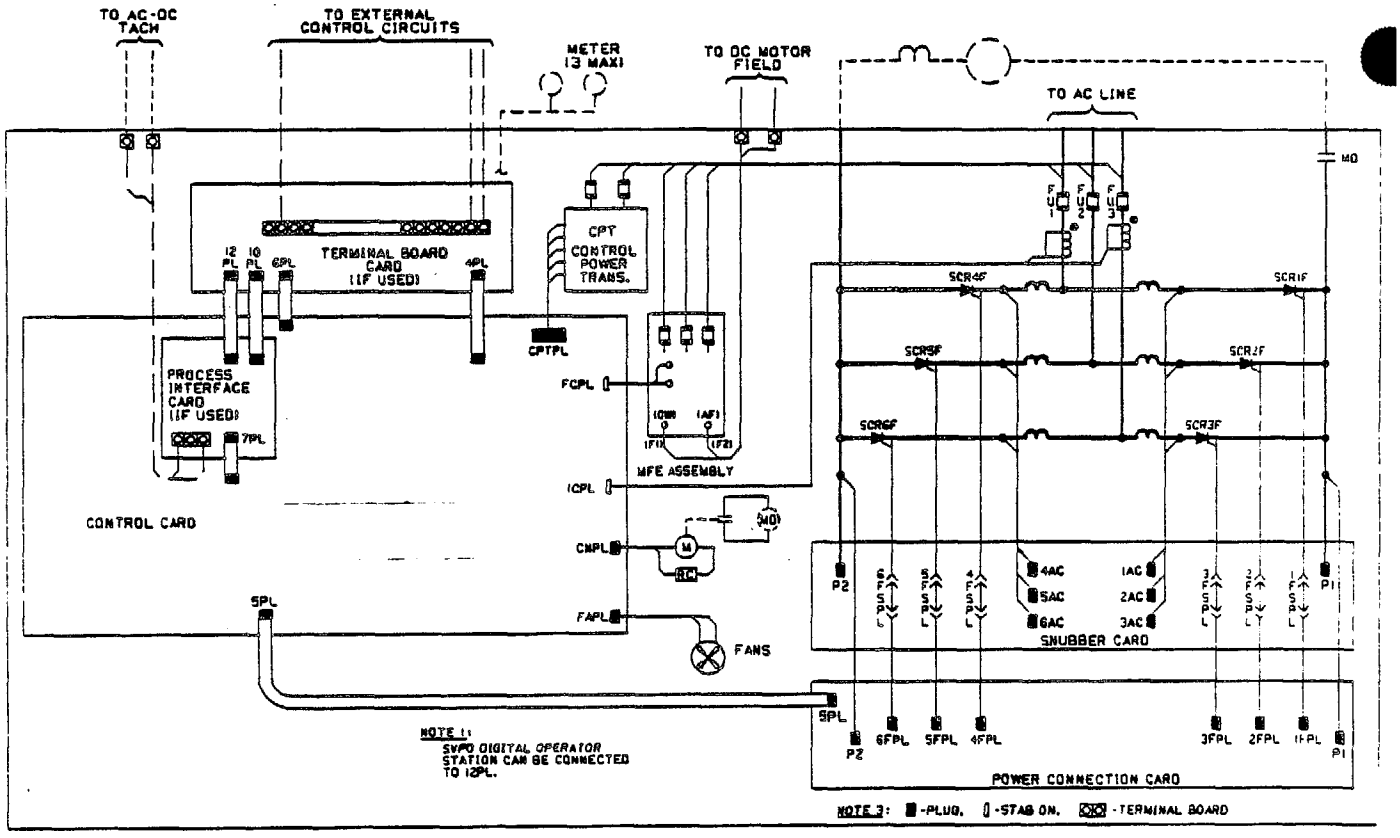
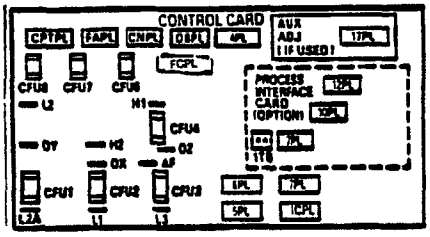
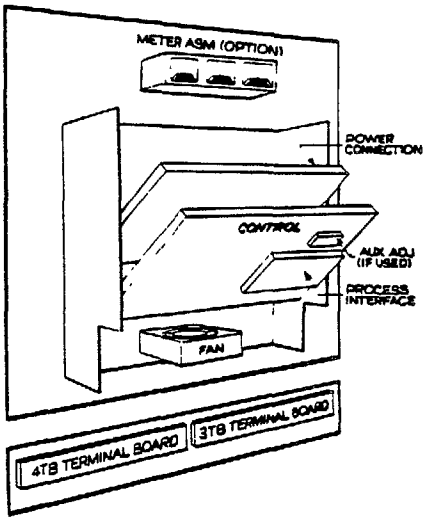


FIGURE 10 — CONNECTION DIAGRAM—NON REGENERATIVE CONTROLLER





For Service

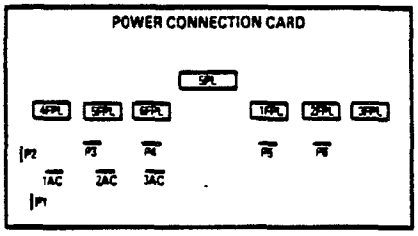
For Factory Service and Technical information on this product call Erie Pennsylvania 814-455-3219 (24 Hour Phone Service)-or contact the nearest GE Apparatus & Engineering Services Office listed in your telephone directory Before calling list Model and Serial numbers of the power unit and motor

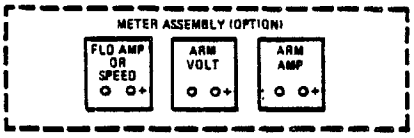
For Parts

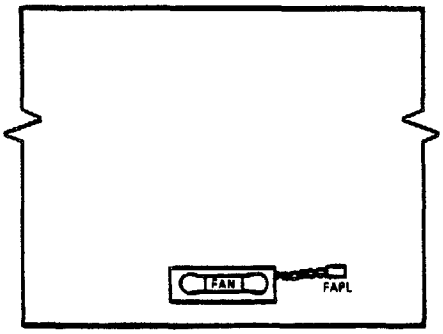
Component parts for this equipment can be obtained from your nearest GE Apparatus Sales representative or authorized General Electric Company distributor

If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures request your representative to contact the Drives Products Operation Erie Pennsylvania giving Model number ML number and Serial number of the complete equipment

Additional information can be obtained from the equipment instruction book



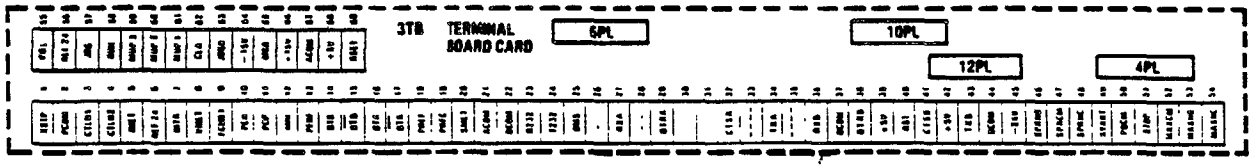


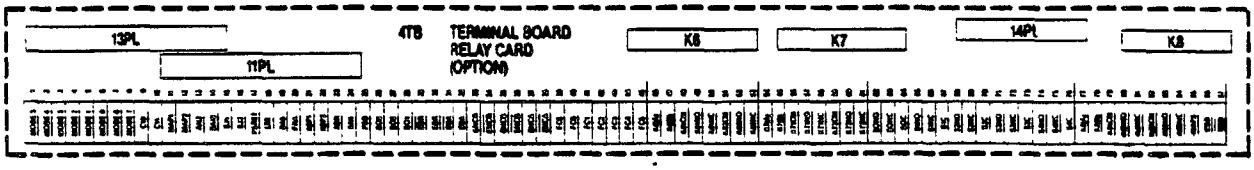


MOTOR FIELD CONTROL: POWER SUPPLY FUSE

Vac	NAME	A	V	TYPE
230/460	CFU1 3	15	600	ATM*
230/460	CFU4	3.2	500	FNQ**
230/460	CFU6	2 0	250	312***
230/460	CFU7	0 5	250	312***
230/460	CFU8	1 0	250	312***

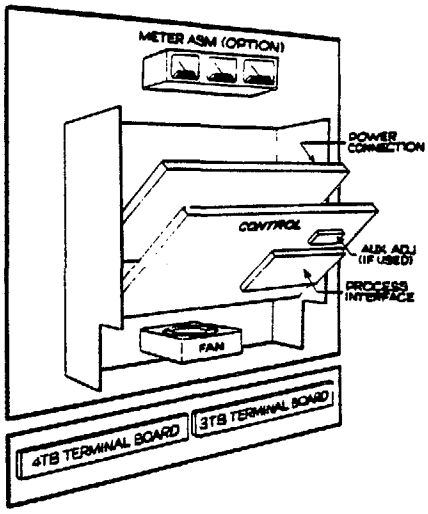
*Gould Shawmut or Equivalent
**Bussmann or Equivalent
***Little Fuse or Equivalent





NP 104X9058A466 2-86

FIGURE 11
DC-285 INFORMATIVE LABEL
NON-REGENERATIVE DRIVE CONTROLLER



For Service

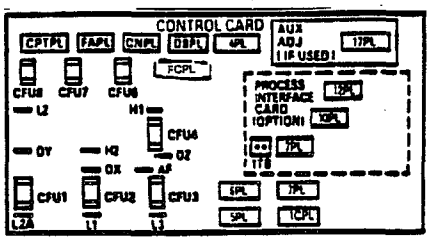
For Factory Service and Technical information on this product call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service)-or contact the nearest GE Apparatus & Engineering Services Office listed in your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor

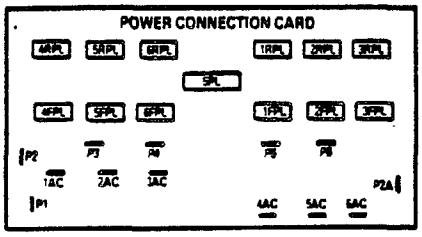
For Parts

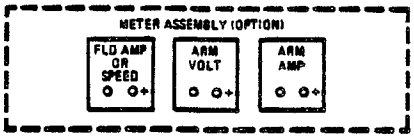
Component parts for this equipment can be obtained from your nearest GE Apparatus Sales representative or authorized General Electric Company distributor.

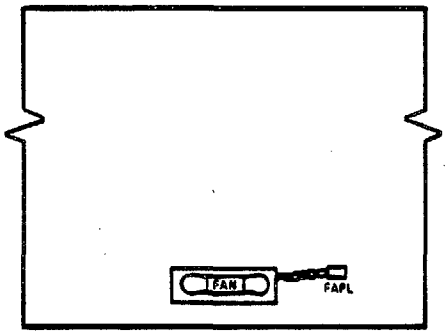
If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures request your representative to contact the Drives Product Operation Erie Pennsylvania giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.



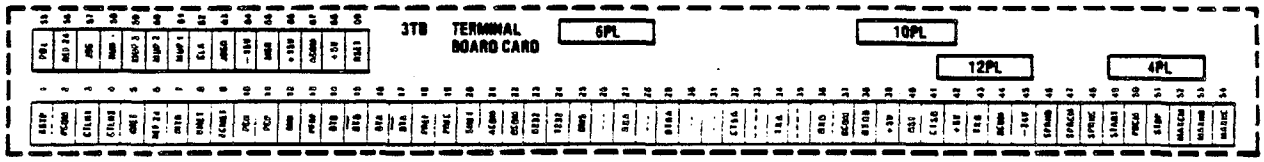


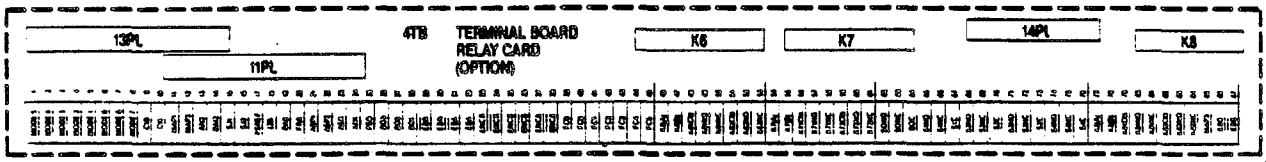




MOTOR FIELD CONTROL POWER SUPPLY FUSE				
Vac	NAME	A	V	TYPE
230 / 460	CFU1-3	15	600	ATM*
230 460	CFU4	3.2	500	FNQ**
230 / 460	CFU6	2.0	250	312***
230 / 460	CFU7	0.5	250	312***
230 / 460	CFU8	1.0	250	312***

*Gould-Shawmut or Equivalent
**Bussmann or Equivalent
***Little Fuse or Equivalent





NP 104X9058A467 2/86

FIGURE 12
DC-285 INFORMATIVE LABEL
REGENERATIVE DRIVE CONTROLLER

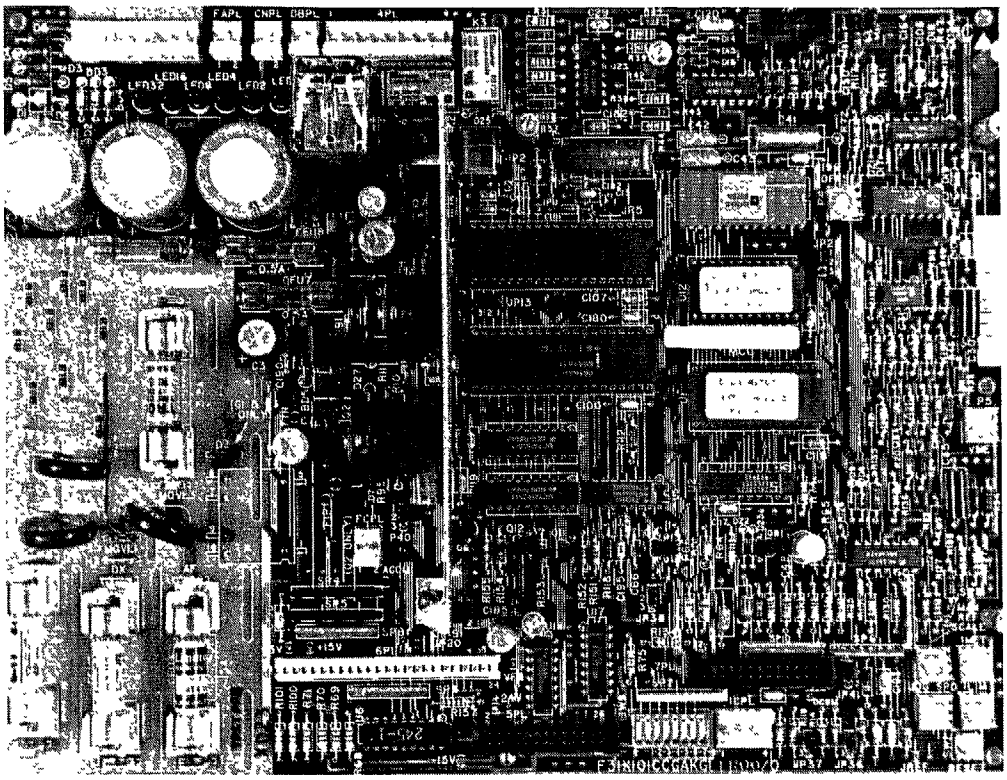


FIGURE 13
DC-285 DRIVE CONTROL CARD

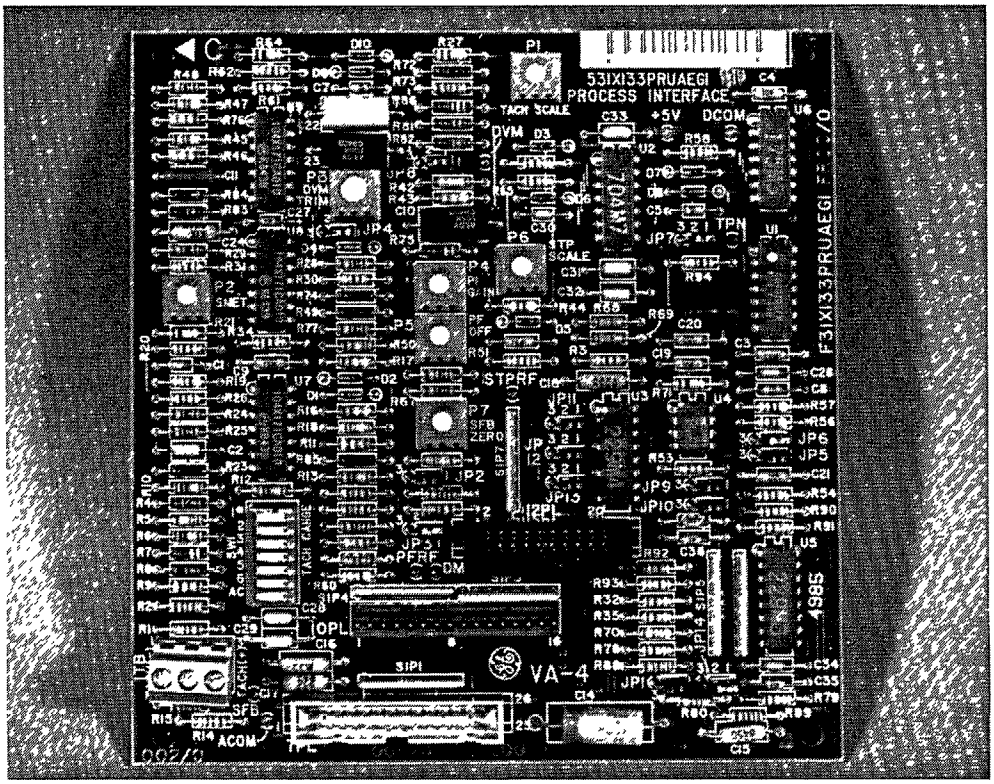


FIGURE 14
OPTIONAL PROCESS INTERFACE CARD

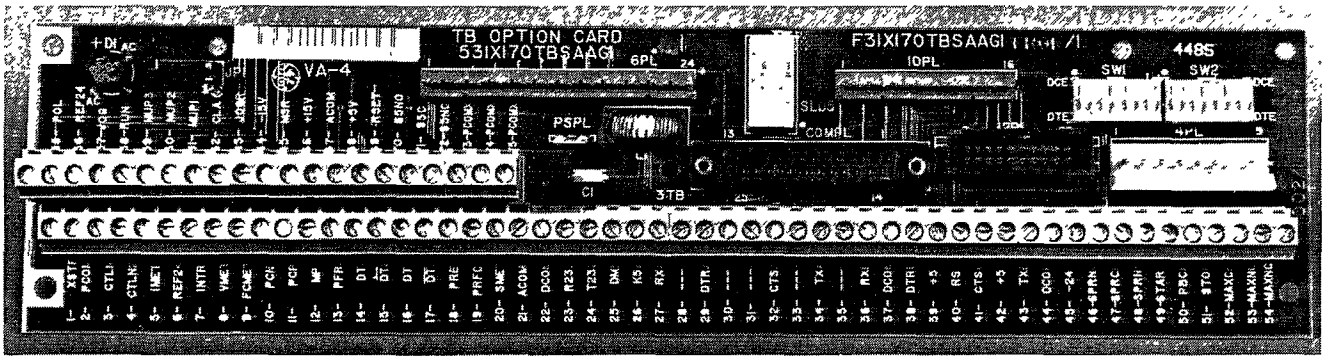


FIGURE 15
TERMINAL BOARD CARD

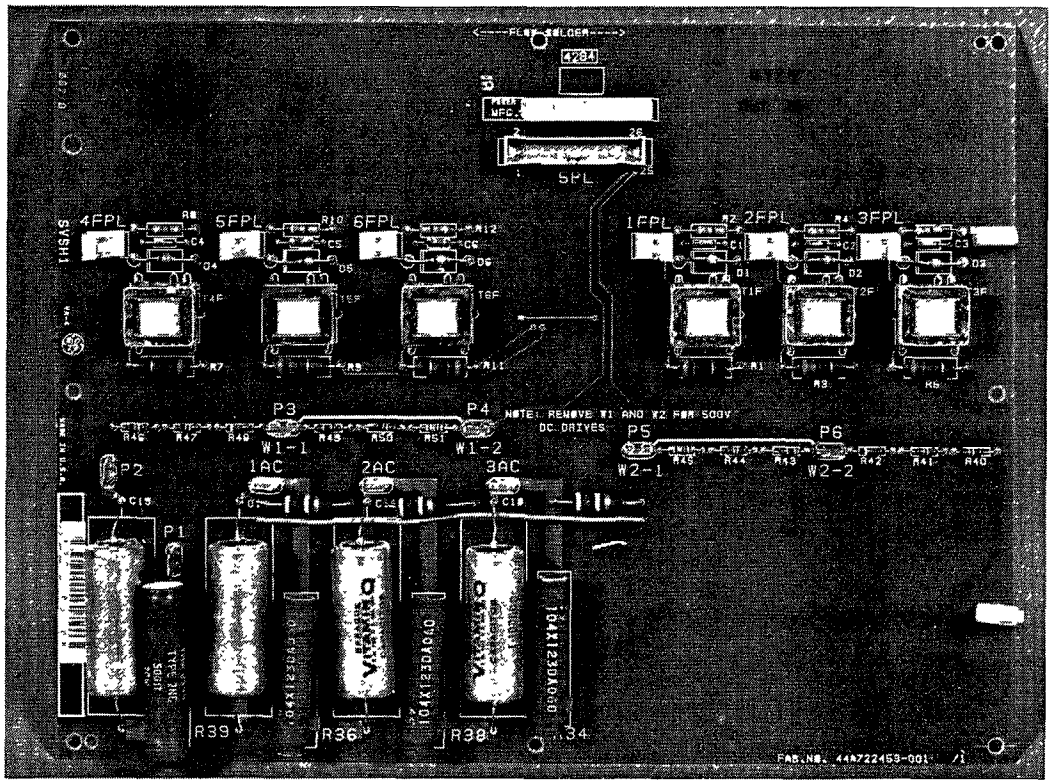


FIGURE 16
POWER CONNECTION CARD
NON-REGENERATIVE DRIVE CONTROLLERS

CONNECTIONS

POWER CONNECTIONS

The three phase input AC power cables are connected at L1, L2 and L3 on the line side of the AC fuses.

The DC motor shunt field leads are connected at 6TB4 (+) and 5 (-).

The DC motor armature cables are connected at P1 (+)at the top of the contactor and P2 (-) at the top of the DC fuse (FU4) and/or buss as shown in Figure 2.

CONTROL CONNECTIONS (See Figures 9 & 10)

The control wiring to the basic controller is terminated at 3TB on the Terminal Board card for all Control card and most Process Interface card connections. Points are referenced by both nomenclature and number. Additional connections may be made through 6TB:

6TB POINT CONTROL FUNCTION DESCRIPTION

1, 2, 3	230/460V, 3 phase input, used when non-standard input voltages are supplied at L1, L2, L3
4	Shunt Field Excitation (+)
5	Shunt Field Excitation (-)
6	External Fault Interlock
7	External Fault Interlock
8	Optional 115 VAC (.25kVA) output
9	Optional 115 VAC (.25kVA) output
10	SCR stack thermal switch output (Stack No. 1)
11	SCR stack thermal switch output (Stack No. 1 and 2)
12	SCR stack thermal switch output (Stack No. 2)
13	Tachometer Feedback input (+)
14	Tachometer feedback input (—)
15, 16, 17, 18, 19, 20	MD Contactor Auxiliary Interlock: (15 to 16 = N.O.) (16 to 17 = N.C.) (18 to 19 = N.O.) (19 to 20 = N.C.)
21	Optional MOV panel fuse loss interlock
22	Optional MOV panel fuse loss interlock

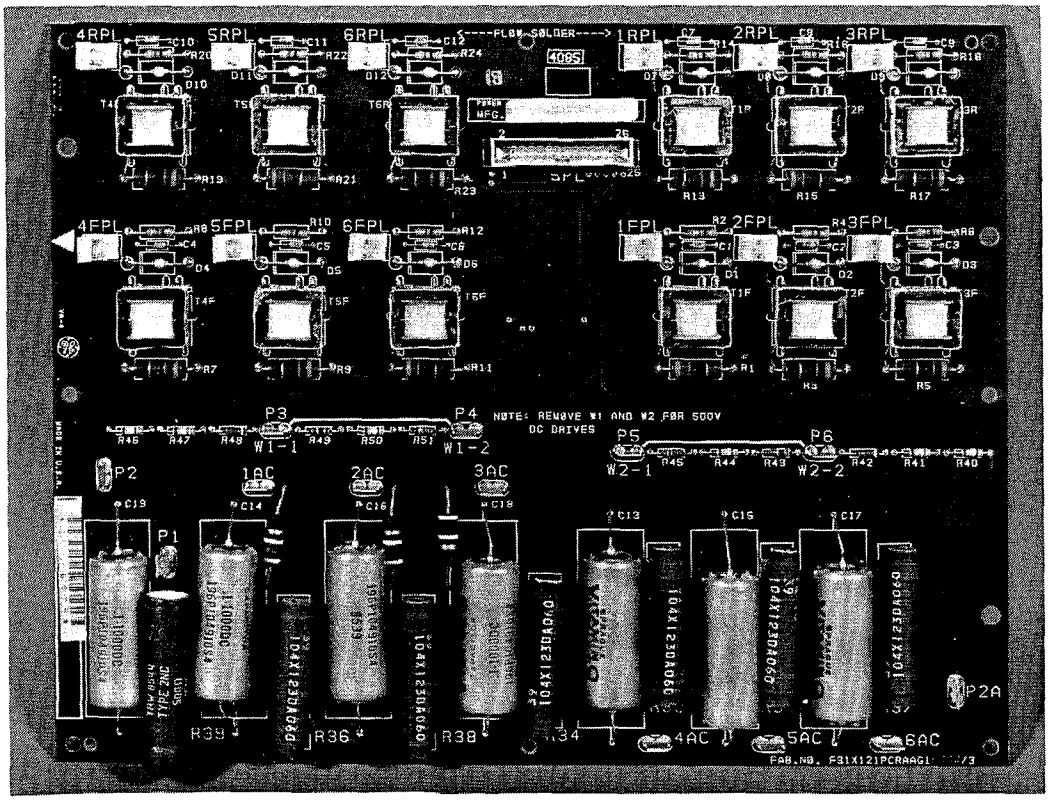


FIGURE 17
POWER CONNECTION CARD
REGENERATIVE DRIVE CONTROLLERS

CONTROL SIGNAL COMMON CONNECTIONS

In those instances where the control connections are brought directly to the circuit boards through 4PL, 6PL, 10PL and 12PL, the following Procedure must be followed with respect to signal commons: ACOM (6PL3) should be used as the common connection for:

- o Analog Signals, References, etc.
- o Serial Link communication signals
- o Drive to Drive coordination

PCOM (6PL, pin 17) should be used as the common connection for meters.

When terminal board 3TB is provided, all signal commons must be connected to PCOMX on 3TB.(73, 74, 75).

MOTOR FIELD EXCITER CONNECTIONS

The DC-285 drive is equipped with a static motor field exciter that provides a fixed, unregulated DC output voltage for developing motor field excitation current. The actual field current is a result of the motor field voltage divided by the field resistance. In most cases this will allow satisfactory motor operation at rated base speed and below.

In some cases, due to low field resistance or high AC input voltage, the resulting motor field currents may exceed the maximum rated base speed field current. It, therefore, may not be possible to achieve rated base speed at full armature voltage. A measurement of the motor field current should be made. If it is found that the field current is above rated maximum it may be necessary to add a trim resistor in series with the field winding. The resistance can be approximated by measuring field current and field voltage at normal operating conditions and using the following calculations.

R(trim) = R(total) - R(field)

Where. . .
R(trim) is approximate resistance to be addeded in series with the motor field.

R(total) is the resulting total resistance required to achieve rated field current.

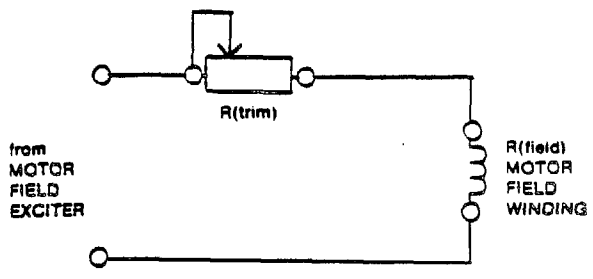
R(field) is the actual field resistance at operating temperature. (Note that this value may be different from the nominal cold resistance given on the motor rating plate)

Using Ohms Law and actual measured current and voltage, R(trim) may be approximated as. . .

$$R\ (trim) = \frac{\text{measured V (field)}}{\text{rated I (field)}} - \frac{\text{measured V (field)}}{\text{measured I (field)}}$$

Where the rated I (field) is the required maximum field current for base speed operation.

Using an adjustable resistor will allow the field current to be fine tuned. The Resistor should be sized per the maufacturers recommendations for the maximum field current and voltage.



START/STOP/ CIRCUITRY OPERATION

The only allowable locations for Control card jumpers 19 & 20 (JP19 & JP20) are in the 1-2 and 3-4 positions respectively.

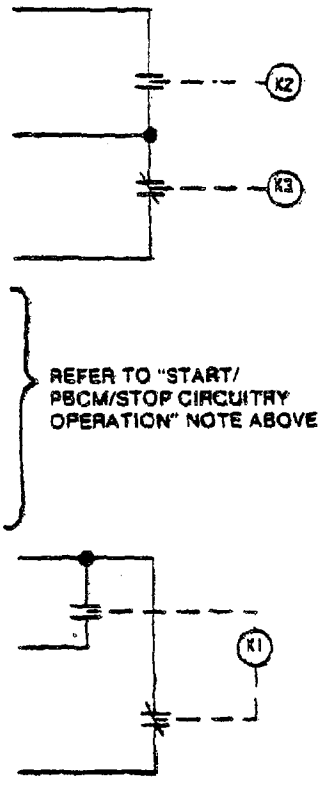
With the jumpers in this position, and with STOP connected to PBCM (or -24V), the START input is active (allows controller to run) when connected to PBCM (or-24V). When START is released, the controller will continue to run until STOP is released from PBCM (or -24V). STOP may then be returned to PBCM (or -24V) in preparation for the next start command.

If START is permanently connected to PBCM (or -24V) then a controller run will be initiated when STOP is connected to PBCM and will stop when STOP is released from PBCM.

The function of each connector and its pins is described in the following:

CONNECTOR 4PL — LOCATED ON CONTROL CARD

4PL PIN NO.	3TB NOMENCLATURE	DESCRIPTION
1	SPRNO (46)	Up to Speed Normally Open Contact
2	SPRCM (47)	Relay Common Connection
3	SPRNC (48)	Fault Relay Normally Closed Contact
4	START (49)	Connection Point for Start Pushbuttons
5	PBCM (50)	Common Point for Start and Stop Pushbuttons
6	STOP (51)	Connection for Stop Pushbutton
7	MAXCM (52)	MA Auxiliary Relay Common Connection
8	MAXNO (53)	MA Auxiliary Relay Normally Open Contact
9	MAXNC (54)	MA Auxiliary Relay Normally Closed Contact



Note: Refer to elementary diagrams for contact ratings.

CONNECTOR 6PL - LOCATED ON CONTROL CARD

PIN NO.	3TB NOMENCLATURE	DESCRIPTIONS
1	RSET (69)	A hardware (hard) reset of drive is initiated when RSET is connected to +15V and must be open to run. RSET immediately stops firing of all SCRs, drops delayed RSET SCR firing power, MD contactor, fault relay, and holds the microprocessor in a reset state until RSET is opened. The drive will not auto matically restart if reset while running. Do not apply RSET while running. Do not apply RSET while regenerating.
2	+5V (68)	Not used. Do not use external to controller
3	ACOM (67)	Not used. Do not use external to controller
4	+15V (66)	Signal voltage for System Reference pot and RESET circuit.
5	MSR (65)	Manual system reference input to drive ± 15V max. positive for forward motoring with POL at zero or open.
6	-15V (64)	Signal Voltage for system reference pot.
7	JOGR (63)	Jog reference input (bypasses linear time), ±14.4V = Top speed if JOG SPD potentiometer is in the full CW position.
8	CLA(62)	Current limit adjust-adds to setting of current limit. +10V increases current limit by about 125% of set,-10V decreases it by approximately 125%, depending on ILIM pot setting.
9	MUP1 (INC) (61)	Control input that commands INCrement speed in the MRH mode.*
10	MUP2 (DEC) (60)	Control input that commands DECrement speed in the MRH mode*
11	MUP3 (DA8) (59)	Eight bit digital to analog current feedback signal. 2 volts = 100% current.
12	RUN (58)	Must be activated (pulled to REF24) to enable controller run. When released, a coast and trip stop is initiated.
13	JOG (57)	Jog command input point. Activated by connecting JOG to REF24. Controller stops when JOG is released.
14	REF24 (56)	±24V supply for XSTP, JOG, POL, REV, INC, DEC or INTR input.
15	POL (55)	Control input which reverses the direction of the drive for any applied reference; e.g, activating POL is equivalent to reversing the sign of the input at MSR. Activated connecting POL to REF 24.
16	XSTP (1)	Must be pulled to REF24 to enable controller. When released, a coast and trip stop is initiated.
17	PCOM (2)	Return line for meter drivers IMET and VMET.
18	CTLN1 (3)	Must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor causing an uncontrolled stop. If the MA contactor is opened under load, harmful contact arcing may occur.

* To activate the MRH mode, the "INC" input must be pulled to REF24 while START is pulled to PBCM. Both inputs can then be released. The controller will continue to run and obey the "INC" and "DEC" commands until STOP is released from PBCM.

CONNECTOR 6PL — LOCATED ON CONTROL CARD (continued)

PIN NO.	3TB NOMENCLATURE	DESCRIPTION
19	CTLN2 (4)	See CTLN1.
20	IMET (5)	Analog load meter output proportional to armature current. See Appendix II for details on meter kits.
21	REF24 (6)	±24V supply for RUN, JOG, POL, XSTP, INC, DEC or INTR.
22	INTR (7)	SLD5 control signal.
23	VMET (8)	Analog armature voltmeter output. This output is 4 volts at rated armature volts output. Maximum load is 2 milliamps.
24	FCMET (9)	Not used.

TERMINAL BOARD 1TB LOCATED ON OPTIONAL PROCESS INTERFACE CARD

1	TKN	Negative tachometer feedback signal.
3	TKP	Positive tachometer feedback signal.

CONNECTOR 10PL — LOCATED ON OPTIONAL PROCESS INTERFACE CARD

10PL PIN NO.	3TB NOMENCLATURE (#)	PIN NO. DESCRIPTION
1	PCN (11)	Negative input } from transducer, 1-5, 4-20, 10-50, 5-65MA or from 0-10V voltage follower
2	PCP (10)	Positive input }
3	MR1 (12)	Reference for manual system reference potentiometer to allow coordinated max speed scaling when PFRF is connected to Control card speed reference input via an auto/manual switch.
4	PFRF (13)	Process follower interface output (0-10V) which connects to a speed reference or other analog input.
5	Not used (14)	
6	Not used (15)	
7	Not used (16)	
8	Not used (17)	
9	Not used (18)	
10	Not used (19)	
11	SMET (20)	Analog speed meter output.
12	ACOM (21)	Common for optional RS232C communications.*
13	DCOM (22)	Not for use external to the controller.
14	R232 (23)	RS232C receive line.
15	T232 (24)	RS232C transmit line.
16	DM5 (25)	Dummy point 5 connects to undedicated point DM5.

* When Terminal Board 3TB is provided, all signal commons must be connected to PCOMX on 3TB (Terminals 73, 74, 75).

CONNECTOR 12PL — LOCATED OPTIONAL PROCESS INTERFACE CARD

12PL PIN NO.	3TB NOMENCLATURE(#)	PIN NO. DESCRIPTION
1	—	No connection
2	—	No connection
3	—	No connection
4	—	No connection
5	—	No connection
6	—	No connection
7	CTSA (32)	Inverting RS422 clear-to-send input from serial link signaling when the Control card may transmit a byte (active high). (EIA standard nomenclature is CTSA).
8	DTRA (29)	Inverting RS422 data-terminal-ready output from Control card signaling when it is ready to receive a byte on the serial link (active high). (EIA standard nomenclature is DTRA).
9	TXA (34)	Inverting RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXA).
10	RXA (27)	Inverting RS422 recieved data signal from serial link to Control card. (EIA standard nomenclature is RXA).
11	RXB (36)	Non-inverting RS422 received data signal from serial link to Control card. (EIA standard nomenclature is RXB).
12	TXB (43)	Non-inverting RS422 transmitted data signal from Control card to serial link. (EIA standard nomenclature is TXB).
13	DTRB (38)	Non-inverting RS422 data-terminal-ready output from Control card to serial link. (EIA standard nomenclature is DTRB).
14	CTSB (41)	Non-inverting RS422 clear-to-send input from serial link to Control card. (EIA standard nomenclature is CTSB).
15	RST (40)	Reset signal from Control card, active high.
16	+5V DC (39)	+5V DC supply — do not use external to the controller.
17	+5V DC (42)	+5V DC supply — do not use external to the controller
18	DCOM (37)	Not for use external to the controller.
19	DCOM (44)	Not for use external to the controller.
20	—24V (45)	Not for use external to the controller.

START—UP PROCEDURE

(REVIEW WARNINGS AND CAUTION NOTICES BEFORE PROCEEDING)

Every DC-285 DC SCR drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are followed.

(If the Hand-Held Programmer is connected through a Process Interface card, refer to instruction book in the Appendix II.)

The DC-285is equipped with a 2/3 wave motor field exciter circuit. No internal adjustments can be made to vary the field strength. Field loss detection circuitry is included and is enabled by placing jumper JP40 in the 1-2 position. Field loss detection can be disabled by placing jumper JP40 in the 2-3 position.

1. Verify that all terminal board screws are tight.

2. Verify that the incoming power is the proper voltage and frequency and the incoming wiring is complete and correct. Also insure that the motor armature, field and other options (i,e., trim resistor) are properly connected.

3. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the following requirements; also confirm that all other jumpers are in accordance with the jumper selection chart supplied with the drive.

3.1 Control Card: (See Figure 13)

SW9-1 MAN REF RANGE	Manual Master Speed Ref. Scaling Opened Position: 5-8V maximum Closed Position: 8-15V maximum

4. Verify that the DC motor has been properly installed and its shaft is free to rotate.

5. Apply power to the drive.

6. If the diagnostic LEDs on the Control card are not blinking one at a time from right to left, press and release the RESET switch on the Control card.

   If the LEDs do not begin blinking, refer to fault codes and recommended action as provided in the Troubleshooting section.

7. Check for proper direction of rotation of blower motor (s) on the DC motor (s). Also when drive is greater than 600HP, 460V, check rotation of drive blower motors.

8. Check the field for proper voltage. Since the drive is in field economy until a "START" is called for, the field voltage will be approximately 105 VDC for a 230 VAC input and 210 VDC for 460 VAC input.

9. Set the external speed reference potentiometer to zero and press the "START pushbutton. The MD contactor should pick up. The diagnostic LEDs should begin blinking two at a time from right to left. The field voltage should increase to 300 VDC for 460 VAC input or 150 VDC for 230 VAC input.

10. Insure that all "STOP" buttons are functional, dropping out MD. Press the "START" button again.

11. Slowly increase the speed reference until the motor starts to rotate. If motor does not rotate, check the diagnostic LEDs. If DC Tachometer Feedback is provided in the system, the most probable cause is an incorrect tach polarity and the LEDs illuminate as follows:

000101
where 0 represents LED OFF
and 1 represents LED ON

Stop the drive. Remove power, interchange the tach leads and repeat.

12. Increase the speed reference potentiometer CW and check the motor rotation. If incorrect, remove power and interchange the motor field leads. If a DC tach is provided in the system, interchange the tachometer leads at this time.

Most of the pots and jumpers are used in the interface circuits for the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use (See Table IV for complete listing and description of these jumpers, pots and switches.) The Custom Software Description (Figure 18) shipped with each controller indicates the jumper positions and EEROM values as they were set in the factory (it does not indicate pot positions.) Pot positions and other adjustments not set in the factory can be made by referring to the information contained in Table IV.

NOTE: Only high impedance digital voltmeters should be used to make indicated measurements.

13. Restart the drive and slowly turn the speed-pot to mid-position (50% speed). Adjust the MAN REF SCALE potentiometer (P1) to obtain 2 VDC at SRS on the Control card. (Measure with a volt-ohmmeter between the appropriate test post and ACOM.) Check actual speed by measuring tach output voltage or use handy tach.

14. **Gradually apply maximum reference: adjust "MAN REF SCALE" potentiometer (P1) on the Control card until 4 volts is read at post SRS on the Control card. If required, change position SW9-1 (right most switch in the DIP switch package) on the Control card to change full scale reference range).**

CAUTION

CONTROLLER MUST BE STOPPED WHEN CHANGING SWITCH OR JUMPER POSITIONS.

15. Confirm good motor commutation (absence of sparking) under load.

16. Slowly decrease speed. Verify stability through the full speed range. If the drive is unstable under load, using the Handheld Programmer, go to PARAMETER MODE. Vary addresses 015 (BSLPG) and 016 (BSLIG) by INC/DEC buttons until the drive is stable. Also check address 037 (SLEAD) for too high a value.

17. After the drive is stable throughout the full speed range, confirm that armature current does not exceed nameplate rating.

18. For motors rated 16 amps or less, jumpers JP36 and JP37 are both in the 1-2 positions. For motors rated above 16 amps, both jumpers are in the 2-3 position.

19. To set up the proper current feedback scaling, Parameter 035 (CFBSF) must be set according to the proper equation per the following table (see example below).

HORSEPOWER				JP36 JP37 POSITION	MOTOR RATED CURRENT	VOLTAGE AT HS or CS POINT
230VAC	460VAC	575VAC 630VDC	575VAC 700VDC			
125 - 150	300	—	400	2-3	440-525	A
200 - 250	400 - 800	400-1000	500-100	2-3	600-1285	B

A = 5.714 [1 - (80/RATED)]

B = 5.714 [1 - (128/RATED)]
RATED = MOTOR ARMATURE CURRENT (AMPS)
Example:

A 500 HP motor (460 VAC) with rated current of 830 amps requires a voltage 'B' of 5.714 [1-128/830] or 4.83 volts at test post HS (or CS).

Listed below are several suggestions on how to improve drive behavior.

Motor Commutator Sparking or Poor Commutation

Check parameter 024 (IRTLM). Reducing the value of this parameter will slow down the rate of rise of armature current with respect to a step input reference.

Too Much Coast Time After a Regenerative Stop

Check parameter 048 (RSTOP). Reducing this setting will cause the motor to decelerate to a lower speed before MA drops out. However, if RSTOP is set too low a fault (FL35) may result when trying to start.

Other

If low current limits are required, check setting of parameter 040 (ILMMN). No current limits from any other source can be lower than this setting. In other words, set this slightly lower than the lowest current limit commanded from other sources external to the Hand-Held programmer.

Parameter 129 (IRATD) may be set in amperes at a value of 100% amperes if desired to read amperes instead of % on the Hand-Held programmer.

KEY POINTS

CONTROL CARD

VFB = ±4V at 100% Armature Volts
SRS = ±4V at 100% Speed Reference

+15V Power Supply = +15VDC (±5%)
-15V Power Supply = -15VDC (±5%)
+5V Power Supply = +5VDC (±5%)

PROCESS INTERFACE CARD (option)

SFB = ±4V at 100% Speed

NOTE: These voltages should be measured using high impedance meters (20K ohms/volt or higher).

HARDWARE JUMPER AND POT ADJUSTMENT

Most of the jumper selections and pot adjustments have been factory selected and adjusted. **When one of the cards has to be changed make sure the replacement card switches, jumpers and pots are set to be the same as on the replaced card.**

Hardware jumpers and pots are located on the Control card and optional Process Interface card.

NOTE: Only a high impedance digital voltmeter should be used to make indicated measurements.

TABLE IV

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control Card	Switch	SW9-1	MAN REF RANGE	Used in conjunction with P1 (MAN REF SCALE) to scale MSR input. Open for reference voltage 5—8V maximum. Close for reference voltage 8—15V maximum.
		SW9-2	TCH/CMF	Not used.
		SW9-3	DC/AC	Not used.
		SW9-4	SCR TEST	Used to perform SCR cell tests. To request SCR test one time only, close SW9-4, press RESET, open SW9-4 and press START. The test will be run obce and the main contactor will drop out. Test passing will be indicated by Fault 42 indication; failure will be indicated by an SCR fault. Clear the fault by pressing Stop. Normal Start/Stop operation will occur from then on.
		SW9-5	REG/$\overline{\text{NON}}$	Not used.
		SW9-6	REV/$\overline{\text{NON}}$	Not used.
		SW9-7	STALL TRIP EN	Not used.
		SW9-8	ILOOP	Not used.
		SW2	RESET	This switch may be momentarily pushed in order to reset the microprocessor and its associated I/O expander chip.
	Pot	P1	MAN REF SCALE	Used to scale the MSR input. This pot provides scaling by a factor as determined by SW9-1 status: SW9-1 open: 0.44 to 0.82 SW9-1 closed: 0.26 to 0.67
		P2	IMET SCALE	Used to adjust SRS to 4VDC at 100% reference. Used to adjust the output to the load voltmeter circuit. 8 volt output equals 100% of rated current.
		P4	OFFSET	Adjusted by manufacturer, not to be adjusted in the field.
		P19	ILIM	Used to scale the CLA input.
		P20	JOG SPD	Used to scale the JOGR input.
	Jumper	JP2 JP5 JP38		1-2: Should be in this position for proper operation. 2-3: Only for manufacturer's use.
		JP19 JP20		JP19 JP20 1-2 3-4 :Makes REF 24 —24 volts 1-3 2-4 :Makes REF 24 +24 volts

TABLE IV (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control Card	Jumper	JP39		Phase Loss Detector Jumper 1-2 for 440 to 480 VAC 2-3 for 220 to 240 VAC
		JP21 JP22 JP23 JP24		Program memory configuration jumpers. Connect 1-2
		JP36 JP37		Scaling for IL1 and IL3. Set at 1—2 for motors rated < = 16 amps. Set at 2—3 for motors rated > 16 amps.
		JP40		Field Loss Enable Connect 1—2 for field loss detection and trip. Connect 2—3 to disable this function.
		JP13		1-2 disable this function; write is protected enables write; write is unprotected
Process Interface Card (optional)	Pot	P1	TACH SCALE	Used to scale the tach circuit to obain 4.0 volts at test post SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit from 4 volts at maximum motor speed. Maximum current load on this signal is 2 mA.
		P3	DVM	With DVM post connected to +5V post and optional Programmer card set to DVM operation, adjust P3 until Handheld Programmer card displays 5.00.
		P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position, and perform the following procedure: — Apply the current reference corresponding to minimum speed. — Adjust PF OFF for 0 volts at test post PFRF. — Adjust the current reference corresponding to maximum speed. — Adjust PF GAIN for 10.0 volts PFRF. Repeat as necessary. **Note:** To prevent reversing when the current reference is less than the minimum current, the drive should be setup in a non-reversing mode.

TABLE IV (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Process Interface Card (optional)	Pot	P6	STP SCALE	Set point scale, used to adjust scaling of definable input at test point STPRF. Refer to elementaries and engineering notes if used.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. 1-2: Enables RS422 Reception 2-3: Enables RS232C Reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2 JP3** 1-2 1-2 5-65mA current loop follower 1-2 2-3 5-50mA current loop follower 2-3 1-2 4-20mA current loop follower 2-3 2-3 1-5mA current loop follower or 10V voltage follower
		JP4		JP4 selects the speed meter (SMET) circuit mode. 1-2: Unipolar speed meter drive mode. 2-3: Bipolar meter driver mode (DC TACH only).
		JP5 JP6 JP7		Not used
		JP8		JP8 selects analog tach filter mode. 1-2: enables 4MS filter in analog tach feedback; required for an AC tach operation. 2-3: disables 4MS filter; required for speed regulator performance above 20 radians (DC Tach only)
		JP11 JP12 JP13 JP14		**JP11 JP12 JP13 JP14** 1-2 1-2 1-2 1-2 RS422 Serial Link - Standard 2-3 2-3 2-3 2-3 RS232C Serial Link - Standard ($\overline{\text{DTR}}$ low) 2-3 2-3 1-2 2-3 RS232C Serial Link ($\overline{\text{DTR}}$ hi) 1-2 2-3 2-3 2-3 RS422 Xmit/RS232C Hand-shake (Series 6 Programmable controller compatible)

TABLE IV (continued)

<table>
<tr><th>LOCATION</th><th>JUMPER OR POT</th><th>NUMBER</th><th>NOMEN-CLATURE</th><th>DESCRIPTION</th></tr>
<tr><td>Process Interface Card (optional)</td><td>Switch</td><td>SW1</td><td>TACH</td><td>These switches provide coarse scaling of the tach circuit as follows:
<table>
<tr><th>SWI POSITION NO. "CLOSED" OR "ON"</th><th colspan="2">TACH VOLT AT MAX SPEED
DC TACH AC TACH</th></tr>
<tr><td>All "Open" or "Off"</td><td>10 - 12.5V</td><td>11.1 - 13.9V</td></tr>
<tr><td>1</td><td>12.5 - 15.5V</td><td>13.9 - 17.2V</td></tr>
<tr><td>2</td><td>15.5 - 19V</td><td>17.2 - 21.1V</td></tr>
<tr><td>1,2</td><td>19 - 22V</td><td>21.1 - 24.4V</td></tr>
<tr><td>3</td><td>22 - 25V</td><td>24.4 - 27.8V</td></tr>
<tr><td>1,3</td><td>25 - 30V</td><td>27.8 - 33.3V</td></tr>
<tr><td>1,2,3</td><td>30 - 37.5V</td><td>33.3 - 41.7V</td></tr>
<tr><td>4</td><td>37.5 - 46V</td><td>41.7 - 51.1V</td></tr>
<tr><td>1,2,4</td><td>46 - 56V</td><td>51.1 - 62.2V</td></tr>
<tr><td>1,2,3,4</td><td>56 - 70V</td><td>62.2 - 77.8V</td></tr>
<tr><td>5</td><td>70 - 87V</td><td>77.8 - 96.7V</td></tr>
<tr><td>2,3,5</td><td>87 - 105V</td><td>96.7 - 117V</td></tr>
<tr><td>1,2,4,5</td><td>105 - 120V</td><td>117- 133V</td></tr>
<tr><td>1,2,3,4,5</td><td>120 - 144V</td><td>133 - 160V</td></tr>
<tr><td>6</td><td>144 - 180V</td><td>160 - 200V</td></tr>
<tr><td>1,2,4,6</td><td>180 - 205V</td><td>200 - 228V</td></tr>
<tr><td>5,6</td><td>205 - 250V</td><td>228 - 278V</td></tr>
<tr><td>1,2,3,4,5,6</td><td>250 - 310V</td><td>278 - 344V</td></tr>
</table>
NOTE: SW1 position 7 must be off (or open) for DC tachs, on (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.</td></tr>
</table>

TEST POSTS

As a further aid to troubleshooting the DC-285 drive controller, a series of test posts have been provided to allow for specific signal observation and/or measurements. Test posts not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POSTS

TEST POST NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common for analog circuit power supplies.	
DCOM	Common for digital circuit power supply.	
M15V	-15V DC	-15V DC ± 5%
P15V	+15V DC	+15V DC ± 5%
P5V	+5V DC	+5V DC ± 5%
SRS	Scaled speed reference signal	4 volts at top speed
VFB	Armature voltage feedback signal. Positive when forward motoring. Maximum external loading is 1mA	4 volts at rated armature voltage

OPTIONAL PROCESS INTERFACE CARD TEST POSTS

TEST POST NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common of analog circuit power supplies.	
DCOM	Common of digital circuit power supply.	
+5V	+5V DC power supply	+5V DC ± 5%
DVM	Input probe connection for digital voltmeter. Requires optional Programmer.	
SFB	Analog speed feedback signal. No value when digital tach or no tach is being used the drive system.	4 volts at top speed
PFRF	Output of process follower interface circuitry.	10 volts at top speed
DM5	Dummy point 5, connects to 10PL-16.	
TPN	AC tach pulse output.	

NOTE

Controller software can provide either a low current or a high current power bridge SCR self check. Normally the controller is configured for the low current test. Depending on various factors (such as design overload conditions, line voltage, motor armature resistance, inductance, DC Power lead length and wire size, etc.) the low current test may not provide sufficient current to verify that the SCRs are functional. This would result in erroneous diagnostic fault indication of faults FL65 through FL86 and FL97 through FL99. It may be necessary to configure the controller for the HIGH current test. Conversion of the decimal value in address 012, as follows will reveal which test has been implemented.

As an illustration, if address 012 were to have the decimal value 1606, dividing that value successively by 2048, 1024, 512 and 256 and noting if the resultant is less than one determines which test has been implemented.

1606/ 2048 = 0 with a remainder of 1606
1606/ 1024 = 1 with a remainder of 582
582/ 512 = 1 with a remainder of 70
70/ 256 = 0 with a remainder of 70

Division by 256 reveals the status of the test on single motor systems.

The "0" means the LOW current test was implemented,
a "1" would mean the HIGH current test was implemented.

It is unlikely but not entirely impossible that even the HIGH current test may not provide sufficient current to verify the SCRs as functional. In the event this situation is encountered, assistance may be obtained from GE DRIVE PRODUCTS OPERATION, ERIE, PA (Telephone [814] 455-3219).

TROUBLESHOOTING

The DC-285 drive controller detects and displays most of the likely faults that could occur. It displays the faults with a series (interpreted as a binary number) of LEDs on the Control card or with a (name and number) fault message on the optional Handheld Programmer.

The following table lists predefined fault codes in order by the assigned fault number. LEDs are assigned a "bit weight" (based on a power of two) from right to left according to whether the LED is "ON" or "OFF". If any LEDs are "BLINKING", add 64 to the total value to obtain the fault number. For example.

	BINARY NUMBER(s)						
LED INDICATION	**32**	**16**	**8**	**4**	**2**	**1**	**FAULT NUMBER**
Steady Display	0	1	1	1	0	0	16 + 8 + 4 = 28
Blinking Display	0	1	0	1	0	0	16 + 4 + 64 = 84

NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

NOTE: **Blinking lights in succession from right to left are normal operating system indicators. One light blinking from right to left indicates "Ready" to run. Two adjacent lights blinking in pairs from right to left indicate "Running".**

TROUBLESHOOTING GUIDE FOR THE DC-285

It is possible that the Control card or Handheld Programmer may not show a fault but in reality a fault does exist. Before replacing anything the following checks are recommended:

1. Is power applied to the drive and is it the correct voltage?
2. Check the following and replace any open fuses (see fuse location and nomenclature on label inside controller door).
   FU1, 2, 3 (Power Fuses)
   FU4, (Regenerative Controller only—DC Fuse)
   CFU1, 2, 3, 4, (High Voltage Fuses)
   CFU6, 7, 8 (Low Voltage Fuses)
3. If the display still does not illuminate, the Control card or Handheld Programmer should be replaced. If the display does illuminate proceed to the appropriate fault display in the following listing.

The faults listed in the table below may be divided into four categories:

L — The fault is latched, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N — The fault is non-latched, i.e., it will clear itself if the cause of the fault no longer exists.

T — The fault trips the drive, i.e., it causes the controller to remove power from the motor. The controller must be reset. All trip faults are latched.

A — The fault is annunciated only, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. Annunciated faults may or may not be latched.

There are several ways to reset the controller. A software reset may be executed from the Handheld Programmer or by a serial command from a device connected to the serial link. A hardware reset may be executed by pressing the reset switch on the Control card, or by momentarily interrupting incoming power to the drive. See also RSET description for connection 6PL, page 17. Most latched faults may be adequately cleared by a software reset. However, some faults should be cleared by a hardware reset. These are indicated by the letter "H". If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
000001	FL01	IOC	Instantaneous DC overcurrent (L,T) Mechanical Problems. SCR firings. Current regulator response too high—reduce gain. Current feedback isolator circuit failure—check 1CPL.
000010	FL02	OVRSP	Overspeed (sensed by analog tach circuitry) (L,T) Tach feedback circuit failure — check 7PL-3, and tach. wiring. Loss of field—check field current. Speed regulator response too high—reduce gain.
000011	FL03	Spare	
000100	FL04	FLOSS	Loss of field current (L,T) Loose connection in field power circuit. Field feedback circuit failure—check connection AF or DZ. Open fuse in field circuit.
000101	FL05	TPOL	Tach polarity incorrect (L,T) Reversed tach connections.
000110	FL06	JMPER	Soft jumper illegally changed while running (L,T) Drive must be stopped to change parameters from addresses 000 to 014.
000111	FL07	RESET	Reset while running (L,T) Drive powered up with JOG closed or RUN and STOP closed — check 6PL-13, and 4PL - 4 & 6. Momentary loss of incoming power while running.
001000	FL08	TOC	Timed overcurrent (L,T) Drive overloaded—check application. Check field current. TOC sense level adjustment too low-check soft adjustment. TOC gain too high—check soft adjustment.
001001	FL09	LNKTR	Serial link trip (L,T) Drive told to trip by serial link—check for external fault.
001010	FL10	LNKFL	Serial link fault (L,T) Serial link communications locked up. Check clear to send handshakes at 12PL-7 and 14.
001011	FL11	Spare	
001100	FL12	Spare	

*NOTE: LED OFF 1 = LED ON 1 = Blinking LED

SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
001101	FL13	EECHK	Soft parameter checksum error or value out of range (L,T,H) Check all parameters for correct value. Checksum not initialized.
001110	FL14	ILOOP	Current loop regulator saturated (L,T) Main contactor not picking up—check pilot relay operation. Loss of current feedback—check 1CPL Armature not connected. DC fuse open. 5PL disconnected.
001111	FL15	VFBLS	Loss of armature voltage feedback (L,T) Motor rotating while no armature volts applied. VFB circuit failure—check test point VFB. SFB circuit failure—check for SFBB zero.
010000	FL16	Spare	
010001	FL17	XSTOP	Coast stop trip (L,T) Check external input at 6PL-16. Check parameter 004 (STPJP).
010010	FL18	STALL	Stalled overcurrent (L,T) Check for mechanical problems. Check field current. Check for settings of stall gain and threshold.
010011	FL19	OVERV	Excessive armature voltage (L,T) Speed or CEMF regulator unstable. Armature voltage feedback circuit failure—Check test point VFB.
010100	FL20	TLOSS	Loss of Tachometer feedback (L,T) Check external tach connections. Tach feedback circuit failure—check 7PL-3 & 13. Process Interface jumpers or switches incorrect.
010101	FL21	NCTRP	Trip from normally closed input on 6PL-22 (RUN) Check connection to 6PL-12 and/or 3TB-58. Check external input at 6PL-12.
010110	FL22	PHLOS	AC phase loss (L,T) Fuse blown in AC line. Low AC line or momentary line notch.
010111	FL23	Spare	
011000	FL24	U3 FL	Bad Control card (L,T) Replace the Control card.

* NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
011001	FL25	COMTR	SCR commutation failure (L,T) Low AC line. Incorrect settings of JP36 & 37 on the Control card. Incorrect connection of AC line CTs.
011010	FL26	ACIOC	Circulating AC line fault (Regenerative controller only (L,T) Bad SCR. Low or noisy AC line. Incorrect setting of JP36 & JP37 on the Control card.
011011	FL27	LSYNC	Loss of AC line sync signal (L,T) Check FL22 causes first. Check for 50/60 Hz at test point SY and U4-12 on the Control card. AC line frequency too high or too low.
011100	FL28	RMODE	Illegal selection of run mode (L,T) Do not command DIAGNOSTIC RUN or STATIC while running.
011101	FL29	Spare	
011110	FL30	Spare	
011111	FL31	NOSEQ	Phase sequence indeterminate (L,T) Noisy or bad AC line—check line feedback circuits.
100000	FL32	TOCAN	Timed overcurrent annunciation (N,A) Drive overloaded—check for mechanical problems. TOC sense level adjustment too low—check adjustment. TOC gain too high—check adjustment.
100001	FL33	PHSEQ	Phase sequence reversed (N,A) Negative sequence inhibit soft jumper selected—reverse any two AC lines. Check three phase motor rotations.
100010	FL34	PLLSY	AC line out of sync (N,A) Noisy AC line. Line frequency too high.
100011	FL35	OSPD	Motor rotating when run commanded (N,A) RSTOP soft adjustment too low. High offset in tach processing circuits. Motor driven externally (overhauling load).
100100	FL36	LNKER	Bad message detected on serial link by drive or other unit (N,A) Check for excessive noise on link or bad Handheld Programmer or Digital Operator Station.
100101	FL37	Spare	

*NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
100110	FL38	ELOCK	Attempt to change software adjustment when soft locked (N,A) Reprogram parameter address with proper setting.
100111	FL39	EFAIL	Software adjustment not stored on EEROM (N,A) JP21-24 on the Control card in hard lock position replace Control card.
101000	FL40	MXSPD	Speed reference greater than max. speed adjustment (N.A) Check scaling of reference and MAN REF SCALE (P1 or P10) on the Control card. Check setting of MAX SPEED.
101001	FL41	KEYOF	Keypad function disable (N,A)
101010	FL42	RLOGC	Run logic error (N,A) Check RUN, JOG, and XSTP inputs for mismatch.
101011	FL43	Spare	
101100	FL44	Spare	
101101	FL45	SLOOP	Speed loop tracking fault (L,T) Drive entered current limit or CEMF limit while tracking was enabled. Speed loop error junction cannot be brought into reglation.
101110	FL46	Spare	
101111	FL47	Spare	
110000	FL48	Spare	
110001	FL49	U4BAD	Bad Control card (L,T) Replace the Control card.
110010	FL50	Spare	
110011	FL51	U1ROM	Bad Control card (L,T) Replace the Control card.
110100	FL52	Spare	
110101	FL53	Spare	
110110	FL54	Spare	
110111	FL55	Spare	
111000	FL56	Spare	

* NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
111001	FL57	Spare	
111010	FL58	U5COM	Bad Control card. Replace Control card
111011	FL59	Spare	
111100	FL60	Spare	
111101	FL61	Spare	
111110	FL62	Spare	
111111	FL63	Spare	
000000	FL64	Spare	
000001	FL65	A1FOP	SCR 1F is not operating (L,T) Check for open gate lead/replace SCR 1F See NOTE preceding TROUBLESHOOTING section.
000010	FL66	A2FOP	SCR 2F is not operating (L,T) Check for open gate lead/replace SCR 2F See NOTE preceding TROUBLESHOOTING section.
000011	FL67	A3FOP	SCR 3F is not operating (L,T) Check for open gate lead/replace SCR 3F See NOTE preceding TROUBLESHOOTING section.
000100	FL68	A4FOP	SCR 4F IS NOT OPERATING (L,T) Check for open gate lead/replace TM4 on regenerative drives, TM1 on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.
000101	FL69	A5FOP	SCR 5F is not operating (L,T) Check for open gate lead/replace SCR 5F on regenerative drives, SCR 2F on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.
000110	FL70	A6FOP	SCR 6F is not operating (L,T) Check for open gate lead/replace SCR 6F on regenerative drives, SCR 3F on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.
000111	FL71	FWDOP	All forward SCRs are not operating (L,T) See NOTE preceding TROUBLESHOOTING section.
001000	FL72	ARMOP	High armature voltage detected during SCR test (L,T) Check for open armature circuit or blown DC fuse. See NOTE preceding TROUBLESHOOTING section.

* NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

SPECIFIC FAULT TROUBLESHOOTING (continued)

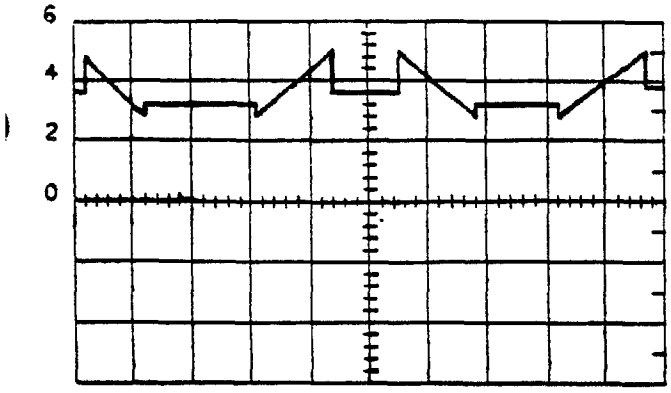
DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
001001	FL73	A1RPO	SCR 1R is not operating (L,T) Check for open gate/replace SCR 1R See NOTE preceding TROUBLESHOOTING section.
001010	FL74	A2ROP	SCR 2R is not operating (L,T) Check for open gate lead/replace SCR 2R See NOTE preceding TROUBLESHOOTING section.
001011	FL75	A3ROP	SCR 3R is not operating (L,T) Check for open gate lead/replace SCR 3R See NOTE preceding TROUBLESHOOTING section.
001100	FL76	A4ROP	SCR 4R is not operating (L,T) Check for open gate lead/replace SCR 4R See NOTE preceding TROUBLESHOOTING section.
001101	FL77	A5ROP	SCR 5R is not operating (L,T) Check for open gate lead/replace SCR 5R See NOTE preceding TROUBLESHOOTING section.
001110	FL78	A6ROP	SCR 6R is not operating (L,T) Check for open gate lead/replace SCR 6R See NOTE preceding TROUBLESHOOTING section.
001111	FL79	REVOP	All reverse SCRs are not operating (L,T) See NOTE preceding TROUBLESHOOTING section.
010000	FL80	SCTFL	SCR short circuit test failed (L,T) Check for multiple power module failures. See NOTE preceding TROUBLESHOOTING section.
010001	FL81	A1FSH	SCR IF or 4R is shorted (L,T) Replace SCR 1F or 4R See NOTE preceding TROUBLESHOOTING section.
010010	FL82	A2FSH	SCR 2F or 5R is shorted (L,T) Replace SCR 2F or 5R See NOTE preceding TROUBLESHOOTING section.
010011	FL83	A3FSH	SCR 3F or 6R is shorted (L,T) Replace SCR 3F or 6R See NOTE preceding TROUBLESHOOTING section.
010100	FL84	A4FSH	SCR 4F or 1R is shorted (L,T) Replace SCR 4F on regenerative drives. SCR 1F on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.

* NOTE: 0 = LED OFF 1 = 1 LED ON 1 = Blinking LED

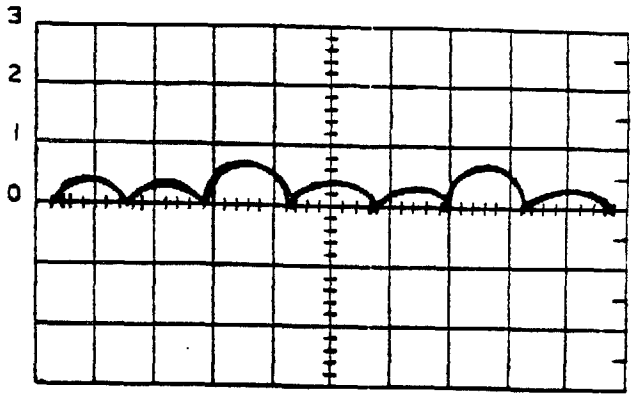
SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
010101	FL85	A5FSH	SCR 5F or 2R is shorted (L,T) Replace SCR 5F on regenerative drives. SCR 2F on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.
010110	FL86	A6FSH	SCR 6F or 3R is shorted (L,T) Replace SCR 6F on regenerative drives. SCR 3F on non-regenerative drives. See NOTE preceding TROUBLESHOOTING section.
010111	FL87	Spare	
011000	FL88	Spare	
110100	FL89	Spare	
011010	FL90	Spare	
011011	FL91	Spare	
011100	FL92	Spare	
011101	FL93	Spare	
011110	FL94	Spare	
011111	FL95	Spare	
100000	FL96	Spare	
100001	FL97	OCTL	SCR open circuit failed (L,T) See NOTE preceding TROUBLESHOOTING section.
100010	FL98	OCTMF	Multiple open SCRs (L,T) Check for open gate leads. See NOTE preceding TROUBLESHOOTING section.
100011	FL99	ALLOP	All SCRs tested open (L,T) Check for Power Connection card not being connected. Check for open DC fuse or armature. Check for contactor operation. See NOTE preceding TROUBLESHOOTING section.

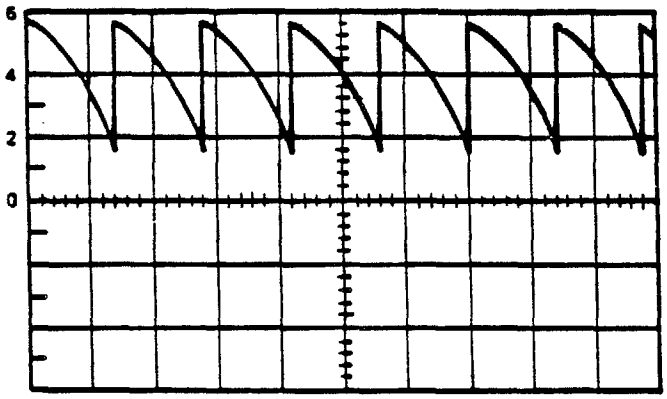
* NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED



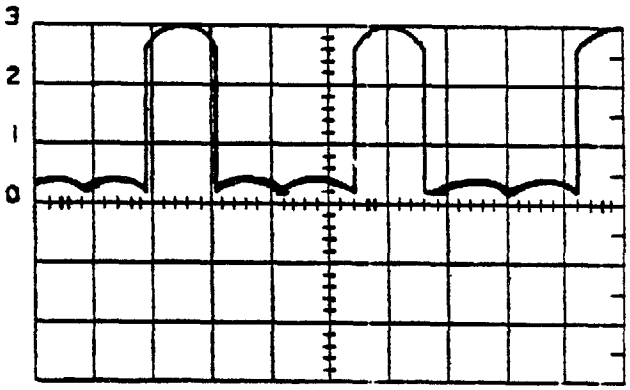
(2MS/DIVISION)
FIGURE 18
VOLTAGE FEEDBACK (VFB) AT
DISCONTINUOUS CURRENT



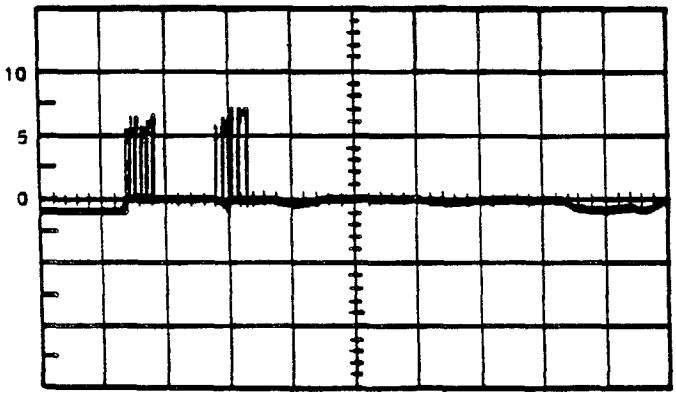
(2MS/DIVISION)
FIGURE 19
CURRENT FEEDBACK (IMET) AT
DISCONTINUOUS CURRENT



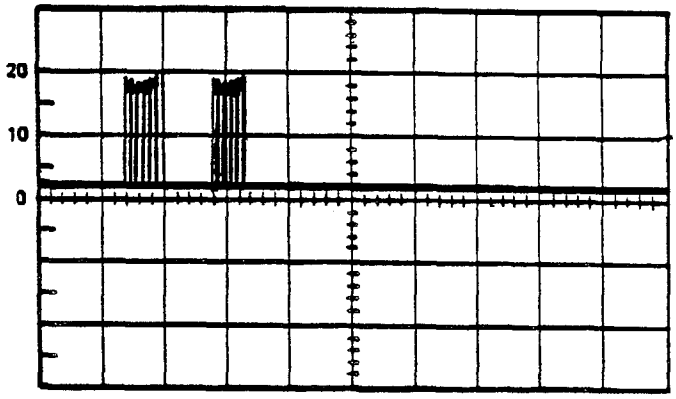
(2MS/DIVISION)
FIGURE 20
VOLTAGE FEEDBACK (VFB) AT
CONTINUOUS CURRENT



(2MS/DIVISION)
FIGURE 21
CURRENT FEEDBACK (IMET) AT
CONTINUOUS CURRENT



(2MS/DIVISION)
FIGURE 22
GATE PULSES WITH GATE CONNECTED



(2MS/DIVISION)
FIGURE 23
GATE PULSES WITH GATE DISCONNECTED

ENCLOSURE DIMENSIONS (see note)

575 VAC			
HP	D"	H"	W"
400 to 800	20	90	48
900 to 1000	20	90	60

460 VAC			
HP	D"	H"	W"
300 to 500	20	90	48
600 to 800	20	90	60

230 VAC			
HP	D"	H"	W"
150 to 200	20	90	48
250 to 300	20	90	60

NOTE: Enclosure dimension W (width) may vary from the tables shown due to optional equipment ordered.
See your Custom Instruction Book for exact dimensions of enclosures provided.

PRINTED CIRCUIT CARDS

DESCRIPTION	CATALOG NUMBER
Control Card	531X101CCGA_M4 or M8*
Power Connection Card (Regenerative)	531X121PCRA_G1
Power Connection Card (Non-Regenerative)	531X122PCNA_G1
Terminal Board Card	531X136TBGA_G1 or 531X170TBSA_G1
Process Interface Card (optional)	531X133PRUA_G1
Digital Operators Station (optional)	531X133PRUA_G1 or 6VHHP1A1
Snubber/Exciter Card	531X124MSDA_G1
SCR Snubber Card	531X126SNDA_G1

*MOVs are mounted on M4 version control cards; they are mounted external of the card on M8 version control cards

SUB ASSEMBLIES AND COMRONENTS (NOT HP RELATED)

DESCRIPTION			CATALOG NUMBER
Motor Field Module			104X125DC012
Control Power Transformer (230V AC)			36B605538AFG01
Control Power Transformer (460V AC)			36B605538AFG02
Suppressor			36A358207AAG01
Fan (small)			104X215CA003
Fan (large)			104X215CA004
Current Transformer Assembly			36A358218YBG02
Fuses (Control Power)	CFU7	0.5 amps	104X109AD009
	CFU6	2 amps	104X109AD085
	CFU4	3.2 amps	104X109BC010
	CFU1-3	15 amps	104X109BC015
	CFU8	0.5 amps	104X109AD009
Fuses (Exciter Assembly)		30 amps	104X109AD084

SPARE AND RENEWAL PARTS (HP RELATED — 230 VAC)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER				
		125	150	200	250	300
AC Power Fuse	104X109AG ___	027	027	030	030	32
DC Power Fuse 4Q only	104X109AG ___	027	028	030	032	034
Armature SCRs	104X125DA ___	089	089	123	137	137
Armature SCRs (with leads)	36A353476AX ___	G01	G01	G02	G03	G03
DC Contactor	104X106AE ___	008	008	008	009	
DC Contactor	104X106MA ___	—	—	—	—	G01
Curr. Xfmer. Asm.	36A358218YA ___	G01	G01	G02	G03	G03

SPARE AND RENEWAL PARTS (HP RELATED — 460 VAC)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER					
		300	400	500	600	700	800
AC Power Fuse	104X109AG ___	027	030	032	032	032	034
DC Power Fuse 4Q only	104X109AJ ___	025	025	025*	025*	025*	026*
Armature SCRs	104X125DA ___	089	123	137	137	137	137
Armature SCRs (with leads)	36A353476AX ___	G01	G02	G03	G03	G03	G03
DC Contactor	104X106AE ___	008	008	009	009		
DC Contactor	104X106MA ___	—	—	—	—	G01	G01
Curr. Xfmer. Asm.	36A358218YA __	G01	G02	G02	G02	G02	G02

SPARE AND RENEWAL PARTS (HP RELATED — 575 VAC, 630 VDC MOTORS)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER 400	500	600	700	800	900	1000
AC Power Fuse	104X109AR _ _ _	027	029	031	033	027*	027*	029*
DC Power Fuse 4Q only	104X109AW _ _ _	006	007	009	006*	006*	007*	007*
Armature SCRs w/leads	36A353476AX _ _	G03	G03	G03	G03	G03	G03	G03
Armature SCRs	104X125DA _ _ _	137	137	137	137	137	137	137
DC Contactor	104X106AE _ _ _	008	008	008	008			
	104X106MA _ _ _					G01	G01	G01

SPARE AND RENEWAL PARTS (HP RELATED — 575 VAC, 700 VDC MOTORS)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER 400	500	600	700	800	900	1000
AC Power Fuse	104X109AR _ _ _	027	027	029	031	032	027*	027*
Armature SCRs w/leads	36A353476AX _ _	G03	G03	G03	G03	G03	G03	G03
Armature SCRs	104X125DA _ _ _	137	137	137	137	137	137	137
DC Contactor	104X106AE _ _ _	007	008	008	008	009		
	104X106MA _ _ _						G01	G01

* 2 per drive required

NOTE: On Regenerative Controllers with paralleled DC fuses, replace both fuses, even if only one is blown.

GLOSSARY OF TERMS

DFP — Delayed Firing Power

EEROM — Electrically Eraseable Read Only Memory

K1 — MAX Relay

K2 — SLDØ Relay

K3 — SLD1 Relay

LED — Light Emitting Diode

MA — Main AC Contactor

MOV — Metal Oxide Varistor

RAM — Random Access Memory

PL — Plug (may be plug or receptacle)

RS232C — Serial Link Communications Standard

RS422 — Serial Link Communications Standard

SCR — Silicon Controlled Rectifier (Thyristor)

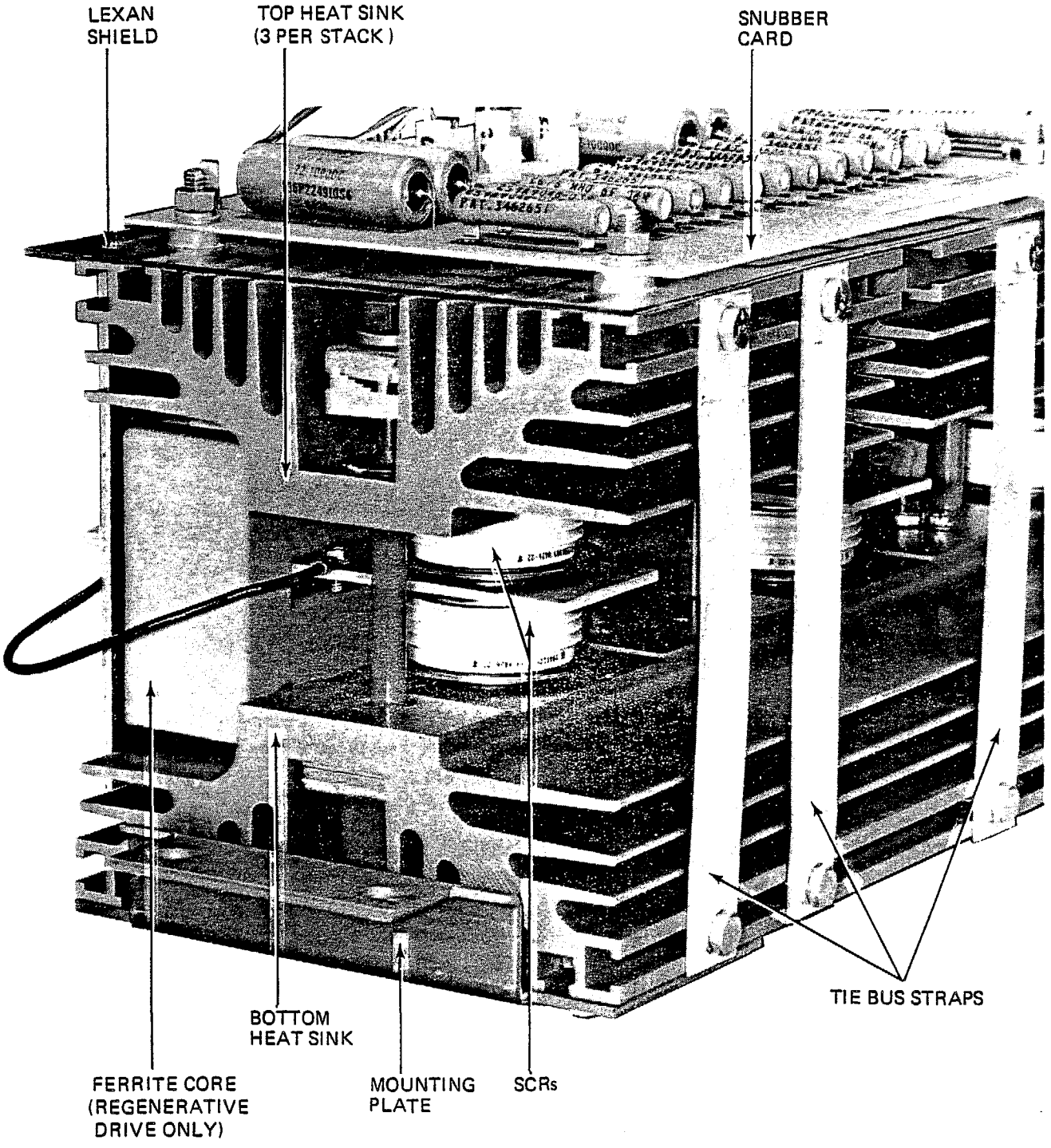


FIGURE 24
DC-285 SCR STACK ASSEMBLY

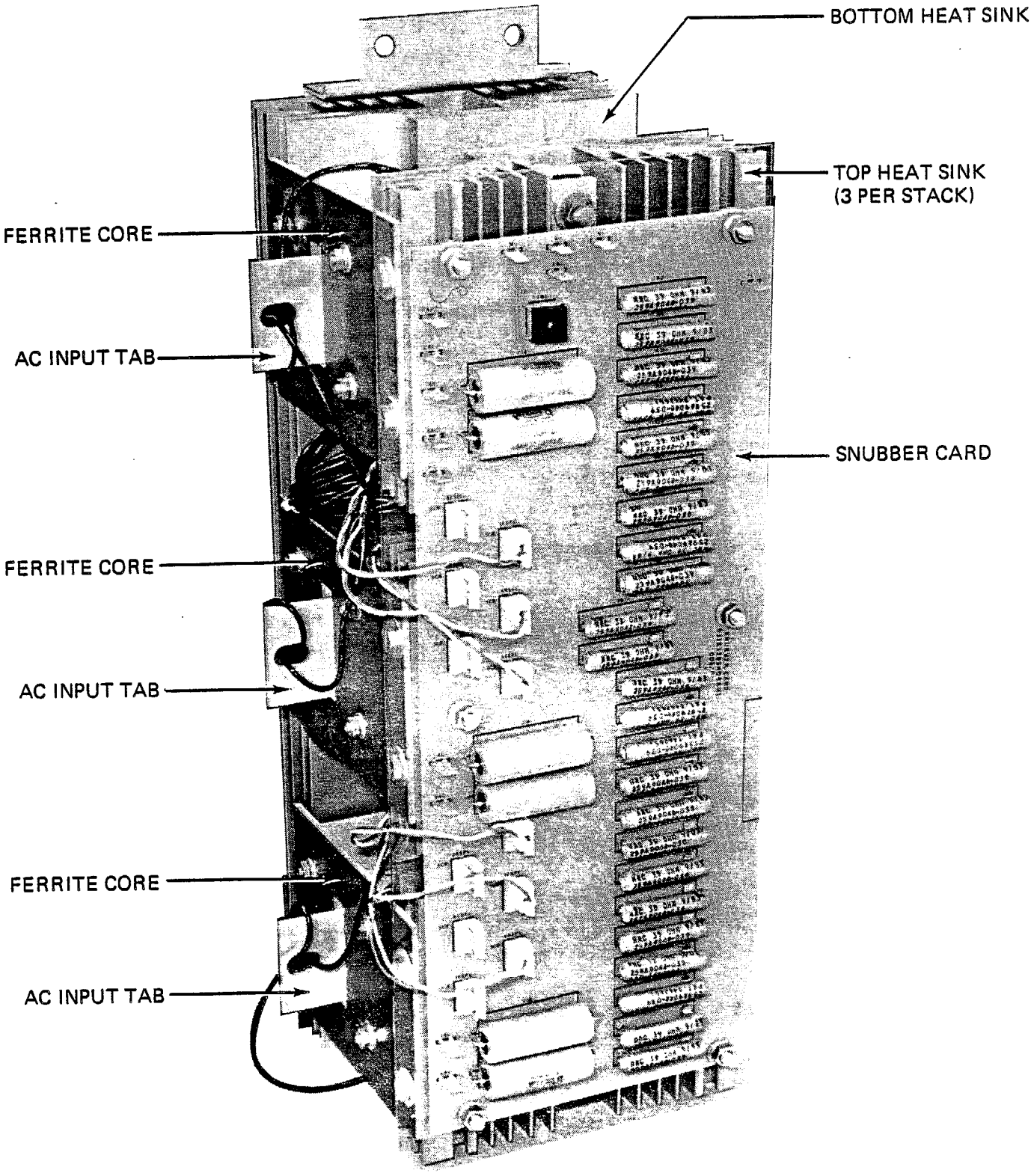


FIGURE 25
DC-285 SCR ASSEMBLY

APPENDIX I
SCR/STACK ASSEMBLY REMOVAL/REPAIR/REPLACEMENT

The following procedure is suggested when it becomes necessary to replace a defective SCR.

Right Hand SCR Stack Removal Procedure

- o Remove all power from the equipment.
- o Remove the SCR firing circuit plugs from the snubber card for the SCRs located in the right hand module only. It is not necessary to remove the SCR firing circuit plugs for the left hand module.
- o Remove the AC wires (1AC__, 2AC__, 3AC__) from the snubber card.
- o Remove the 6 mounting nuts which hold the snubber card in place being careful not to let the bolts drop into the bottom of the controller.
- o Remove the harness retainers from their captive position.
- o Carefully roll the snubber card off the stack and over onto the left hand stack.
- o Remove the three AC line input bolts, again being careful not to lose any hardware.
- o Remove the electrical connection (bus) at the bottom of the stack.
- o Likewise remove the stack mounting bolts (qty 4) at the top and bottom of the stack.
- o The stack assembly can now be lifted off the panel and placed on a workbench for further disassembly.

Left Hand SCR Removal Procedure

- o Follow the same procedure outlined above (except removal of the snubber card), i.e., removal of appropriate SCR firing circuit plugs, 4AC__, 5AC__, 6AC__ leads from the snubber card, mounting hardware, electrical connections.

SCR Replacement Procedure

- o Identify the name of the defective SCR (from fault codes) and SCR location (See Figures 27 and 28.
- o Place the stack assembly on its side and disassemble the section containing the defective SCR.
- o Alternately loosen each nut on the clamp assembly to keep even pressure on the good cell.
- o As each part is removed from the stack assembly, place it on a clean surface and in the same orientation and order in which they were removed.
- o Remove the defective cell. Clean and lubricate the heatsinks before reassembling. Place a few drops of GE SF1153 silicone oil in the center of both heatsink mounting surfaces. Polish the interface area until clean. using a combination of back and forth, sideways, and circular motions with a lint-free cloth or tissue. Finally place a few drops of GE SF1153 silicone oil on the mating surfaces of the heatsink and use a lint-free cloth or tissue to spread the oil evenly. Wipe off excess oil.
- o Clean and lubricate the SCR cell. Place the cell on a flat table and clean the anode and cathode faces with GE SF1153 oil as described above. Apply a thin film of silicone oil to cover the mating surfaces and wipe off any excess.
- o Coat the threads, nuts and washers used in the clamp mechanism with a small amount of graphite spray.
- o Replace the SCR cell carefully, ensuring that its orientation and polarity are correct. Use care when placing the cell on the dowel pin to prevent scratching the cell post surface. Rotate the cell back and forth a few times to ensure proper seating.
- o Reassemble the SCR cell assembly, taking care not to damage the cell with the heatsink. Tighten the clamp nuts only finger tight. Ensure that the number of exposed threads on each stud are about equal.
- o Tighten each nut alternately in accordance with Table I.

TABLE I

SCR Part No.	Number/Turns*	Increments
104X125DA137	1-3/8	1/4 Turn
104X125DA123	3/4	1/4 Turn
104X125DA089	1/2	1/4 Turn

*from finger tight

- o Replace the repaired stack assembly following in reverse the procedure outlined for removal.

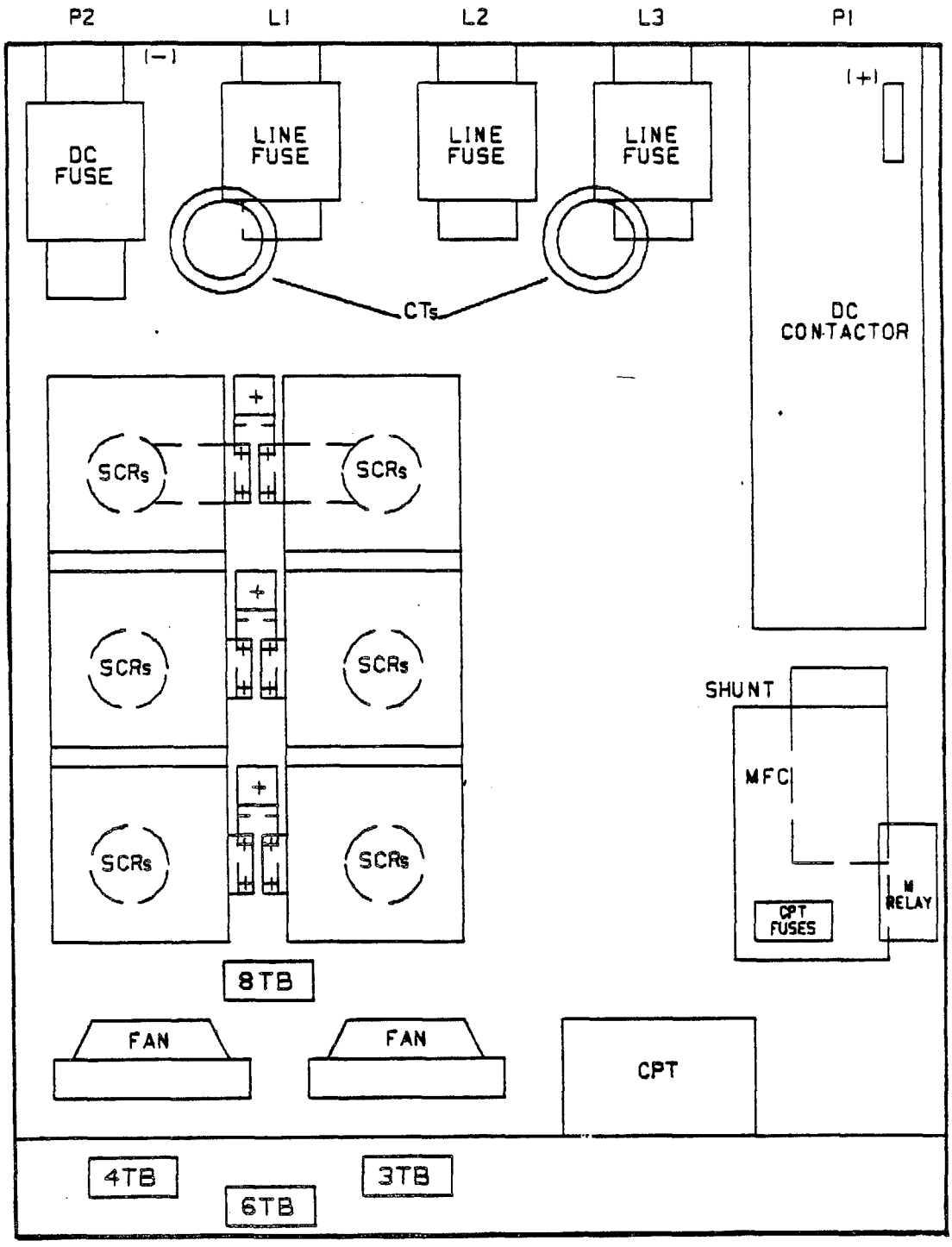
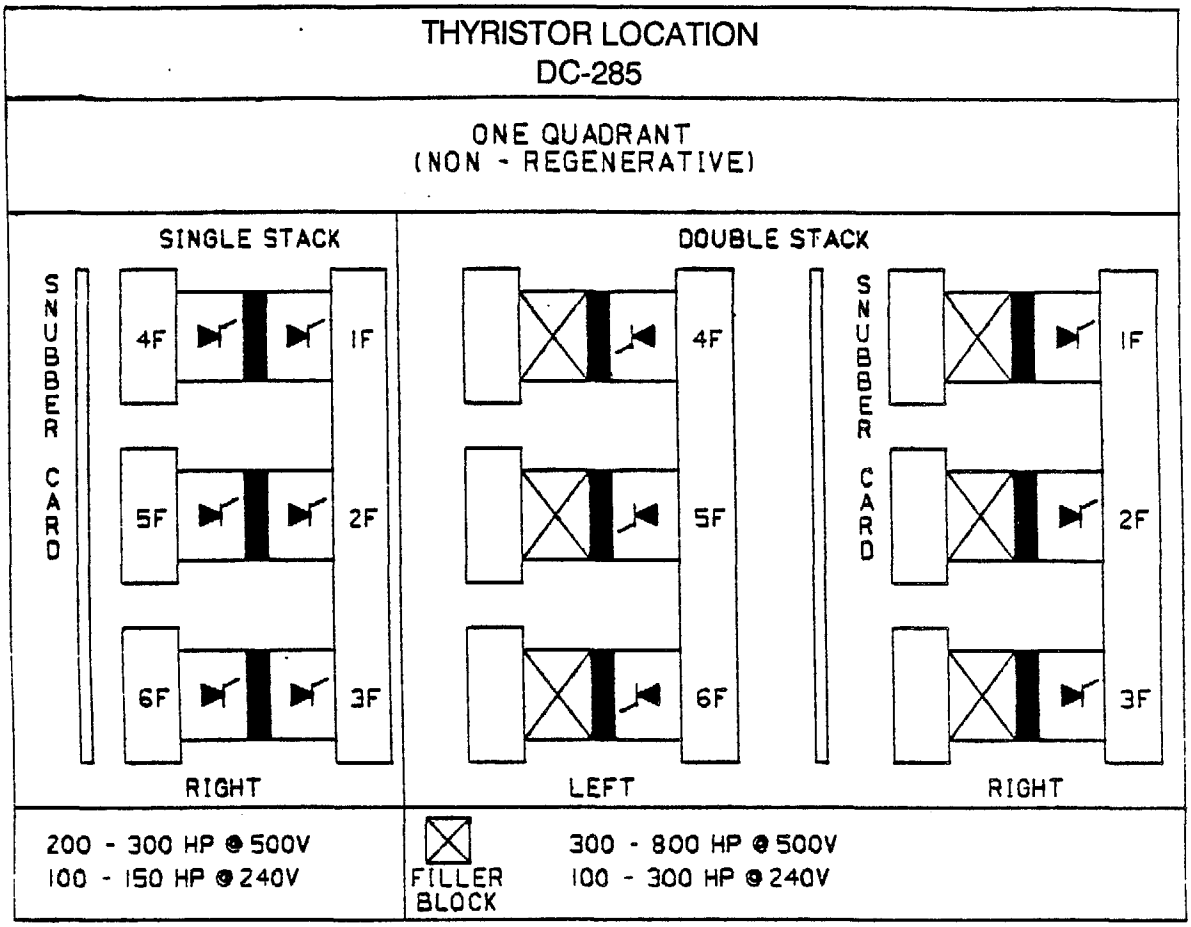


FIGURE 26
SCR LOCATION — GENERAL

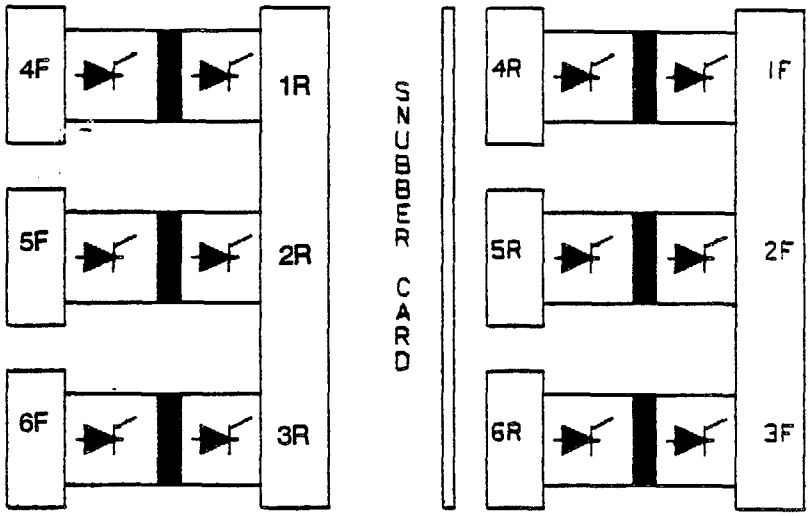


FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*
100 TO 150	FU1-3	500	500	A50P	300	FU1-3	500	500	A50P
					400	FU1-3	800	500	A50P
200	FU1-3	800	500	A50P	500 TO 700	FU1-3	1000	500	A50P
250 TO 300	FU1-3	1000	500	A50P					
					800	FU1-3	1200	500	A50P
* GOULD-SHAWMUT OR BRUSH ELECTRIC OR EQUIVALENT									

FIGURE 27
SCR LOCATION—NON-REGENERATIVE

THYRISTOR LOCATION
DC-285

FOUR QUADRANT
(REGENERATIVE)



300 - 800HP AT 500V
100 - 300HP AT 240V

FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*
100 TO 150	FU1-3	500	500	A50P	300	FU1-3	500	500	A50P
					400	FU1-3	800	500	A50P
200	FU1-3	800	500	A50P	500 TO 700	FU1-3	1000	500	A50P
250 TO 300	FU1-3	1000	500	A50P					
100	FU4	400	500	A50P	800	FU1-3	1200	500	A50P
125	FU4	500	500	A50P	300 TO 400	FU4	700	700	A70P
150	FU4	600	500	A50P					
200	FU4	800	500	A50P	500 TO 700	FU4	700**	700	A70P
250	FU4	1000	500	A50P					
300	FU4	700**	500	A50P	800	FU4	800**	700	A70P

* GOULD-SHAWMUT OR BRUSH ELECTRIC OR EQUIVALENT ** QTY 2 PER DRIVE

FIGURE 28
SCR LOCATION — REGENERATIVE



NOTE: Never plug the programmer directly into
7PL on the 531X101CCGA_M Main Control card.
Connect to 7PL only through adaptor 36A58260RCG01.

FIGURE 29
HANDHELD PROGRAMMER

APPENDIX III

HANDHELD PROGRAMMER '(See Figure 29)'

In addition to the diagnostic displays (See Troubleshooting Section), should a fault occur, the Hand Held Programmer has the following modes of operation.

OPERATE MODE	This mode contains all of the necessary functions to run the drive with preprogrammed speeds or the speed selected by the Handheld Prorgrammer card. The user can also monitor the status of the drive. Indicated faults can be reset.
PARAMETER MODE	This mode allows the user to examine and/or change, if necessary, the software jumpers and adjustments provided in this controller.
DIAGNOSTIC MODE	This mode allows the user to monitor a number of different parameters (for example; the armature voltage or current). This mode also allows for a self test of the controller on the card itself and also place the drive into the diagnostic RUN and STATIC mode.

FUNCTIONS: The following functions should allow most drive operations to be exhibited:

1) RUN
2) SLOW (THREAD)*
3) JOG
4) STOP (CONTROLLED STOP)
5) REVERSE*
6) INCREMENT REFERENCE*
7) DECREMENT REFERENCE*
8) AUTO/MANUAL SELECT KEY*
9) RESET (Software resets the drive)
10) R1 (Auxiliary run command utilizing preset ref. R1RF)*
11) R2 (Auxiliary run command utilizing preset ref. R2RF)*
12) RUN AT REFERENCE
13) RATIO (DRAW)*
14) PROGRESSIVE*
    PROGRESSIVE RUN (FORWARD OR REVERSE)
    SLOW (FORWARD OR REVERSE)
    JOG (FORWARD OR REVERSE)
    STOP
    R1 (FORWARD OR REVERSE)
    R2 (FORWARD OR REVERSE)
    RATIO
15) SET (To observe a drive's reference contents)
16) ENTER (To adjust a drive's reference contents)
17) DRIVE NUMBER (Used along with the SET key in order to key in the drive number 77 which places the user in the PARAMETER MODE).

* Optional functions, not available on all drives.

OPERATING MODE

1) **RUN**

Press the RUN key, to run at RUNRF (address 061) setting speed.

2) **SLOW**

Press the SLOW key, to run at SLWRF (address 062) setting speed.

3) **JOG**

Press the JOG key, to run at JOGRF (address 063) setting speed. (Hold the key down for as long as you want the drive to JOG.)

Pressing a run command (RUN, SLOW, R1, R2 REFR) without pressing the REVerse key first will initiate a forward command. Jog is a special function. When REVerse key is pressed and then the JOG key is pressed and let up, the display will show "JOG REV"
Pressing the JOG key again will again initiate a reverse jog. When the JOG key is let up the display will show "JOG REV".

Pressing a run command (RUN, SLOW R1, R2, REFR) without pressing the REVerse key first will initiate a forward command. However, pressing the JOG key will initiate a reverse jog if the display is showing "JOG REV". To clear the jog function press the RESET key.

Jog progressive and jog regressive work the same way as jog reverse with the exception that the progressive mode may be cleared by pressing the the PRGSV key.

4) **STOP**

Press the STOP key to stop the drive.

5) **REVERSE**

Press the REV key, then press the desired run key (RUN, SLOW, JOG, R1, R2, REFR) to activate any of the reverse running modes.

6) **INCREASE REFERENCE**

Press the INC key. (Hold the key down for as long as you want the motor to increase speed.)

7) **DECREASE REFERENCE**

Same as for increase speed except press the DEC key.

8) **AUTO/MANUAL**

Pressing the A/M key will change the drive from its present mode to its other mode (AUTO MODE TO MANUAL MODE OR MANUAL MODE TO AUTO MODE). The status of the drive will be displayed. The mode can only be changed when the motor is stopped.

9) **RESET**

The RESET key when pressed will send a RESET command to the drive across the serial link, and software reset the Programmer card.

10) **R1**

Press the R1 key to run at reference R1RF.

11) **R2**

Press the R2 key to run at reference R2RF.

12) **RUN AT REFERENCE**

Press the REFR key.

This function will command the drive to run at the referenced speed at which the drive was last running.

13) **RATIO**

Press the RATIO key.
Then press the INC or DEC key until the desired ratio is obtained. Holding the INC or DEC will increment or decrement the ratio twice a second.

14) **PROGRESSIVE XXX** (XXX = RUN, SLOW, JOG, R1, R2, RATIO)
PROGRESSIVE REVERSE XXX (XXX = RUN, SLOW, JOG, R1, R2)

Press the PRGSV KEY. The display will show "**PROGR**".

Press the key that contains the desired function. If the reverse key is the last key pressed, the display will show "**REGRESSIVE**".

Then press a run key (RUN, SLOW, JOG, R1, R2) to initiate the desired function.

If the progressive function is to be performed again, the above procedure will have to be repeated starting with pressing the PRGSV key except if the function is a Jog Progressive function (see JOG).

15 & 16) **SET & ENTER**

The SET key is used to examine and modify any of the run commands and the drive number. (RUN, SLOW, JOG, R1, R2, REFR, DRV#).

Press the SET key.

The display will show "**SET**".

Press the key which contains the function whose contents are to be changed.

The display will show the function's contents.

Press the numbered keys until the desired value is shown on the display. (The INC or DEC keys may be used to increment or decrement the value shown on the display.)

Press ENTER key.

The value on the display has just been entered.

If the value is not to be entered, the SET key may be pressed. The display will show, "**CLR SET**" (After about a second, the drive's status will be displayed and the keypad is in the OPERATE MODE.)

17) **DRIVE NUMBER**

To examine the drive number, press the DRV# key, the operator is still in the OPERATE MODE of the keypad, e.g. the upper functions on the keypad are in effect.

The display will show the drive number and whether or not the drive is present.

"DRIVE 00" Drive number = 00 and Drive 00 is present

"NO DRIVE 00" Drive 00 is not present

To change the drive number, the SET key must first be pressed. (See SET & ENTER.)

PARAMETER ADJUSTMENT MODE

This mode

A) Allows you to examine specific addresses of the DC Controller.

B) Allows you to increment or decrement the contents of the address by incrementing the displayed value and then automatically programming the newly incremented or decremented value into the EEPROM on the controller.

C) Allows you to call up the next address and its contents to the display.

D) Allows you to enter the displayed contents to the displayed address memory in an external device's memory.

E) Allows you to send a controlled STOP command to the drive.

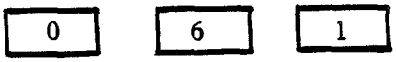
F) Allows you to clear the display and indicate the mode of the keypad in the display and/or change the mode of the keypad.

To get into the software adjustment mode, key in SET DRV #77 and ENTER.

Key a three—digit address using the number keys.

The parameter name and contents will displayed.

EXAMPLE:



Display: **RUNRF. 1500.0**

Use the numbered keys or INC, DEC keys to adjust the displayed value (1500.0). When the desired value is being displayed, press the **ENTER** KEY.

The display will show

ENTR — DRIVE

RUNRF. XXXX.X

XXXX.X is the new value which is stored in the drive's memory.

To display the next parameter in the drive's memory, press the **NEXT** key.

To continue to the DIAGNOSTIC MODE, press the **CLEAR/MODE** key.

To send a **CONTROLLED STOP** command to the drive, press the **STOP** key.

DIAGNOSTIC MODE
This mode

A) Allows you to monitor a number of different parameters by pressing

1) the key with the desired parameters shown, or

2) press the TEST key (the display will show **TEST**; then key in a two-digit number which corresponds to the parameter to be monitored, then press ENTER.

B) Allows the user to initiate a self test to the Handheld Programmer or controller.
Press the TEST key (the display will show **TEST**); then key in a two-digit number which corresponds to the test to be initiated, then press ENTER.

This product will perform the test and show the result on the display.

TEST 51- Will show the revision date of the Handheld Programmer software.
TEST 52- Will show the revision number of the Handheld Programmer's software.

C) Allows the user to place the drive into the diagnostic run or diagnostic static mode by pressing the key labeled DGR or DGS respectively.

DIAGNOSTIC RUN — press the DGR key
To exit from the DIAGNOSTIC RUN MODE, press the STOP key.

DIAGNOSTIC STATIC — press the DGS key
To exit from the DIAGNOSTIC STATIC MODE, press the STOP key.

Once the user is in the DIAGNOSTIC RUN mode, he may change the motor's speed by pressing the INC or DEC keys, or change keypad modes and perform the various functions. To exit from the DIAGNOSTIC RUN MODE, press the STOP key.

D) Allows the user to examine RAM addresses in the controller.
Press the RAM key.
The programmer will display ***RAM***
Enter the desired address using the number keys.
The contents of a five-digit RAM location will then be displayed on the display.
By pressing the CLEAR/MODE key, the user will return to the DIAGNOSTIC MODE.

E) Allows the user to send a controlled STOP command to the drive.

F) Allows the user to change the drive number by:
Press the DRV # key.
The present drive number will be displayed on the display. Key in a two-digit number using the numbered keys. When the desired number is being displayed, press the ENTER key. (If the drive isn't present, the display will show "**NODRIVE XX**" where **XX = THE DRIVE NUMBER**).

G) Allows the user to change the mode of the keypad.
Press the CLEAR/MODE key to continue to the OPERATE MODE.

Below is a list of errors that might be displayed along with the corresponding error number.

NUMBER	NOMENCLATURE	ERROR
	TEST ERRORS	
ER01	BUART	BAD UART ON THE 8031 U.C.
ER02	BKPAD	THE KEYPAD HAS SHORTED LEADS
ER03	BDMEM	THE RANDOM ACCESS MEMORY IS FAULTY
ER04	BPRGM	THE PROGRAM MEMORY CHECKSUM IS INCORRECT
ER05	BTIMR	THE TIMERS IN THE 8031 U.C. ARE FAULTY
	USER ERRORS	
ER06	NADDR	AN ADDRESS IS NOT SPECIFIED
ER07	2KEYS	TWO KEYS WERE PRESSED TOGETHER
ER08	ILBRD	ILLEGAL BROADCAST
	SPECIAL LINK ERRORS	
ER11	NOACK	NO ACKNOWLEDGE RECEIVED
ER12	N—CTS	NO CTS (CLEAR TO SEND)
ER13	ORBUF	RECEIVE BUFFER HAS OVERFLOWED
ER14	CKSUM	INCORRECT CHECK SUM WAS RECEIVED
ER15	BDMSG	SOMETHING OTHER THAN AN S.O.T. WAS RECEIVED WHEN AN S.O.T. WAS EXPECTED.

On the following pages is a listing of DC-285 software adjustments ("Key Parameter Adjustments") common in most drive applications. The parameters can be examined and adjusted with the optional Handheld Programmer (refer to APPENDIX III) or via the serial link.
Additionally, a Custom Software Description is shipped with every drive. The Custom Software Description describes the setting and/or function of various programmable jumpers and adjustments (hardware and software) within the drives that are identified with the specific model number of that drive. Included in that book is a listing of specific programmer options, their description and a set of software adjustment definitions pertinent to the specific drive.

KEY PARAMETER ADJUSTMENTS
The following parameter adjustments, listing Programmer mnemonic, address number, function name, and description, will prove helpful in setup and troubleshooting of the DC-285 drive. This is a partial list. For a more complete list of parameters contained in a specific drive, refer to the Custom Software Description furnished with that drive.

The following is a description of column headers:

NAME	Mnemonic displayed by Programmer when interrogating parameter.
ADR	Parameter address.
UNITS	Unit designation of displayed number (example - RPM).
MIN	Minimum allowable value.
MAX	Maximum allowable value.
STEP	Amount by which value can be incremented or decremented.
DEFAULT	Standard value if no special setting is required.

NOTE: Parameters listed are grouped functionally.

KEY PARAMETER ADJUSTMENTS

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

REFERENCES AND SCALING

REFSF 068 None 0 6553 1 6553
Reference Scale Factor - - sets the factor by which references entered from the serial link and Programmer will be multiplied and by which speeds displayed on the Programmer and sent to the serial link will be divided. Enter the value of speed reference which corresponds to top speed (4 volts at SFB). Important - affects settings of RUNRF, SLWRF, JOGRF, R1RF, R2RF, SLKRF, and speed reference display.

RUNRF 061 RPM 0 6553 1 0
"Run" Key Reference - - sets the preset reference for RUN commands from the serial link. Also may be used as the preset reference for manual start commands. May be a speed, current, current limit or position depending on jumper setting in ADR006-RUNJP.

SLWRF 062 RPM 0 6553 1 0
"Slow" Key Reference - - sets the preset reference for SLOW commands from the serial link.

JOGRF 063 RPM 0 6553 1 10% of REFSF
"Jog" Key Reference - - sets the preset reference for Jog commands from the serial link. Also may be used as the preset reference for manual jog commands. Default should be 10% of REFSF (068).

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

RPMSF 070 RPM 0 6553 1 6553
RPM Scale Factor - - sets the speed in RPM which corresponds to 4 volts at SFB. NOTE: Change parameter only while the drive is not running, and toggle the reset switch afterwards to make the change fully effective. Important - affects setting of REFLM, MINSP, SPDLM, TCLSL, TCLS2, TCLS3, AND TCLSH. (Affects pulse reference scaling.)

REFLM 071 RPM 0 6656 1 6560
Maximum Speed - - sets the level at which speed reference will be clamped if the input reference magnitude exceeds the level. Maximum value should not exceed 1.015 x RPMSF.

MINSP 072 RPM 0 6553 1 0
Minimum Speed - - sets the level at which speed reference will be clamped if the input reference magnitude is less than the level. Maximum value should not exceed RMPSF.

SPDLM 073 RPM 0 7936 1 7208
Speed Limit Trip Setpoint - - sets the speed at which the drive will trip. Should not exceed 1.2 x RPMSF.

ARMATURE CURRENT REGULATOR

ILMMX 041 % Rated 10 510 2 150
Maximum Current Limit - - sets the current limit for speeds below the tapered current limit start speed, TCLSL. One unit = 1% current. Current limit is never less than ILMMN.

ILMMN 040 % Rated 10 510 2 10
Minimum Current Limit - - sets the current limit for speeds above the tapered current limit high speed set point, TCLSH. One unit = 1% current.

IOCTR 043 % Rated 10 510 2 400
Instantaneous Overcurrent Trip Setpoint - - sets the current level at which the drive will trip. One unit = 1% current.

TOCTH 044 % Rated 10 510 2 120
Timed Overcurrent Sense Threshold - - sets the current level at which the TOC integrator begins to time-up. One unit = 1% current.

TOCGN 045 None 0 255 1 24
Timed Overcurrent Response - - sets the time constant of the TOC integrator. Refer to the Custom Software Description if multi-motor function is used.

TOCGN = 66,666 *divided by* (Trip Time) (X-TOCTH)

NOTE: X = % of armature current corresponding to desired trip time.

NOTE: The default values for TOCTH and TOCGN allow the drive to operate at 150% current for 90 seconds.

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

DEADB 023 None 0 255 1 0
Current Regulator Deadband - - controls the sensitivity of the current loop reversal logic. A setting of 0 is maximum sensitivity (dither mode).

IRTLM 024 PU/S 1 255 1 50
Current Rate Limit - - controls the maximum rate of change of current command presented to the current regulators. Exact relation is current rate = 0.9 (IRTLM) per unit per second.

OUTPUT SIGNALS

DAC8 096 None 0 65535 1 0
D-A #1 Output Control - - controls what internal variable is sent to the 8-bit D-A converter (DAC8) on the DC-285 Control Card. See Custom Software Description for complete information.

DIAGNOSTICS

DIAG1 122 None 0 65535 1 0
Diagnostic Variable #1 - - selects 1 of 2 variables which can be displayed on the serial link or Programmer using Test 11. See Custom Software Description for complete information.

DIAG2 123 None 0 65535 1 0
Diagnostic Variable #2 - - selects second variable which can be displayed on the serial link or Programmer using Test 11. See (096) DAC8 output control for programming information.

SCHRL 127 None 1 26 1 18
Left Scaled Unit Character - - determines which character is displayed as the left character of the scaled speed status message. 1 represents "A"; 2, "B"; 26, "Z".

SCHRR 128 None 1 26 1 13
Right Scaled Unit Character - - see ADR127-SCHRL for programming information.

IRATD 129 None 0 3276 1 100
Displayed Arm Current Scaling - - controls the factor by which per unit current is multiplied for display on the Programmer.

NAME	ADR	UNITS	MIN	MAX	STEP	DEFAULT

VRATD **130** **None** **0** **2621** **1** **100**
Displayed Arms Volts Scaling - - controls the factor by which per unit voltage is multiplied for display on the Programmer.

SRATD **132** **None** **0** **6200** **1** **100**
Displayed Speed Scaling - - controls the factor by which per unit speed is multiplied for display on the Programmer.

TRATD **133** **None** **0** **3276** **1** **100**
Displayed Torque Scaling - - controls the factor by which per unit torque is multiplied for display on the Programmer. For field weakened drives, divide the scale factor by the top speed/base speed ratio before entering the value.

HRATD **134** **None** **0** **3276** **1** **100**
Displayed HP Scaling - - controls the factor by which per unit horsepower is multiplied for display on the Programmer.



GE Drives

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GENERAL ELECTRIC COMPANY
DRIVES DEPARTMENT
ERIE, PENNSYLVANIA 16531